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Supporting Information for

# Decision-Relevant Modelling of Extreme Demand for Industrial Electricity Tariffs

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## APPENDIX S1. SUPPLEMENTARY METHODS, DERIVATIONS, AND ANALYSES

This Supporting Information provides technical derivations, implementation details, complete supplementary summaries, robustness analyses, and external predictive analyses referenced in the main article. The primary controlled experiment compares five predictive constructions. TAIL and EVENT-EMP use event-level histories, whereas GEV, KDE, and EMP use the corresponding monthly maxima. The primary controlled experiment contains 10,000 independent 24-month histories and nine prespecified tariff cells, giving 450,000 history–method–cell decisions. Supplementary recovery, stress, physical-clipping, and physical-misspecification experiments are separate experiments and retain their own stated analysis units.

Throughout,  $M \geq 0$  denotes monthly maximum demand,  $F$  its distribution,

$$\mu_F = \mathbb{E}_F[M], \quad H_F(t) = \mathbb{E}_F[(M - t)_+].$$

The reference capacity is  $S$ , the contract level is  $D$ , the contract-threshold multiplier is  $\alpha$ , the lower contract fraction is  $\delta$ , and the exceedance-charge multiplier is  $\kappa$ . Optimized mode values are denoted by  $V_F(m)$ , the oracle value by  $V_F^*$ , oracle mode separation by  $\gamma_F$ , and the maximum mode-value perturbation by  $e_V$ . The retrospective stability ratio is

$$d_{MV} = \frac{2e_V}{\gamma_F}.$$

For a selected full action  $\hat{a}$ , regret is

$$R_F(\hat{a}) = C_F(\hat{a}) - V_F^*.$$

### S1.1 Supplementary tariff identities and decision-stability details

The main article gives the tariff functionals and the principal decision-stability results. This subsection records only the atom-aware and boundary details used by the implementation.

For the contract objective

$$\phi_F(D) = D + \kappa H_F(\alpha D), \quad D \in [\delta S, S],$$

the stop-loss functional has one-sided derivatives

$$H'_{F,+}(t) = -\Pr_F(M > t), \quad H'_{F,-}(t) = -\Pr_F(M \geq t).$$

Consequently,

$$\partial \phi_F(D) = \left[ 1 - \kappa \alpha \Pr_F(M \geq \alpha D), 1 - \kappa \alpha \Pr_F(M > \alpha D) \right].$$

For an interior optimizer this gives

$$\Pr_F(M > \alpha D^*) \leq \frac{1}{\kappa \alpha} \leq \Pr_F(M \geq \alpha D^*).$$

Writing

$$p^* = 1 - \frac{1}{\kappa \alpha},$$

$\alpha D^*$  is therefore a generalized  $p^*$ -quantile. The feasible-boundary cases are handled by restricting the generalized quantile solution to

$$D \in [\delta S, S].$$

*Mode stability and demand-scale radii.*

For a unique oracle tariff mode  $m_F^*$ , define, for each  $m \neq m_F^*$ ,

$$\gamma_{F,m} = V_F(m) - V_F(m_F^*),$$

and, for every  $j \in \mathcal{M}$ ,

$$e_j = |V_{\hat{F}}(j) - V_F(j)|.$$

37 The pairwise sufficient condition is

$$38 \quad e_m + e_{m_F^*} < \gamma_{F,m} \quad \text{for every } m \neq m_F^*.$$

39 With

$$40 \quad e_V = \max_{j \in \mathcal{M}} e_j, \quad \gamma_F = \min_{m \neq m_F^*} \gamma_{F,m},$$

41 the scalar condition

$$42 \quad d_{MV} = \frac{2e_V}{\gamma_F} < 1$$

43 is a simpler conservative sufficient certificate. Both conditions use oracle quantities and are retrospective diagnostics in the  
44 controlled experiment.

45 For the Wasserstein representation, let

$$46 \quad L_{\text{cap}} = 0, \quad L_{\text{act}} = r_d, \quad L_{\text{con}} = r_d \kappa.$$

47 The corresponding analytic demand-scale radii are

$$48 \quad \rho_F^{\text{pair}} = \min_{m \neq m_F^*} \frac{\gamma_{F,m}}{L_m + L_{m_F^*}}, \quad \rho_F = \frac{\gamma_F}{2L}, \quad L = r_d \max\{1, \kappa\}.$$

49 Table S13 reports the primary tariff geometry and these oracle-dependent radii. They contain neither a numerical integration  
50 allowance nor a solver allowance.

51 *Full-action loss.*

52 The exact regret decomposition separates mode-selection loss from within-mode excess. The latter is identically zero for actual  
53 demand and capacity. Only contract demand can have nonzero within-mode excess. If contract demand is both oracle optimal  
54 and selected, and  $\hat{D}$  exactly minimizes the fitted contract objective, then

$$55 \quad R_F(a_{\text{con}}(\hat{D})) \leq 2r_d \kappa \Delta_H(F, \hat{F}),$$

56 where

$$57 \quad \Delta_H(F, \hat{F}) = \sup_{D \in \mathcal{D}_S} |H_{\hat{F}}(\alpha D) - H_F(\alpha D)|.$$

58 As an implementation check, the exact regret decomposition was verified on all 450,000 primary decision rows, with no  
59 failures; the maximum absolute residual was  $7.45 \times 10^{-9}$  CNY/month.

## 60 S1.2 Analytic oracle and deterministic numerical evaluation

61 The controlled data-generating mechanism admits an analytic monthly-maximum distribution. In each month, 200 bounded bulk  
62 events are generated as

$$63 \quad X^{(b)} = 45,000 + 35,000 B, \quad B \sim \text{Beta}(8, 2),$$

64 and an independent number

$$65 \quad N \sim \text{Poisson}(30)$$

66 of tail events is generated as

$$67 \quad X^{(t)} = 80,000 + Y, \quad Y \sim \text{GPD}(\xi = 0.15, \sigma = 12,000).$$

68 The monthly maximum  $M$  is the maximum over the bulk and tail events.

69 Let

$$70 \quad L = 45,000, \quad u_0 = 80,000, \quad \lambda = 30,$$

and let  $F_B$  denote the Beta(8, 2) distribution function. The exact monthly-maximum CDF is

$$F_M(x) = \begin{cases} 0, & x < L, \\ \exp(-\lambda) \left[ F_B\left(\frac{x-L}{u_0-L}\right) \right]^{200}, & L \leq x < u_0, \\ \exp\left[-\lambda \left\{ 1 + \xi \frac{x-u_0}{\sigma} \right\}^{-1/\xi}\right], & x \geq u_0. \end{cases}$$

The resulting oracle quantities are

$$\mu_F = 148,235.371489 \text{ kW},$$

$$q_{0.99} = 265,663.010007 \text{ kW},$$

and

$$q_{0.999} = 375,513.529996 \text{ kW}.$$

The oracle mean and stop-loss functionals are evaluated from

$$\mu_F = \int_0^\infty \{1 - F_M(x)\} dx$$

and

$$H_F(t) = \int_t^\infty \{1 - F_M(x)\} dx.$$

High quantiles are obtained from the analytic distribution. The same oracle law therefore determines the reference capacity

$$S = \frac{\mu_F}{u},$$

the tariff-mode values, the optimized contract level, the unique oracle mode where applicable, and the oracle mode separation  $\gamma_F$ .

Final fitted-law evaluation likewise avoids a common predictive Monte Carlo layer between  $\hat{F}$  and the tariff functionals. Finite discrete distributions use exact sums. The nonnegative Gaussian KDE uses analytic Gaussian-mixture expressions for its mean and stop-loss function. The fitted tail-process and nonnegative-conditioned GEV constructions use deterministic one-dimensional integration where required, and continuous quantiles are obtained by direct inversion or deterministic bracketed root finding.

The deterministic numerical layer was independently checked in 55 numerical comparisons. The maximum relative discrepancy was approximately

$$1.283 \times 10^{-11}.$$

No Monte Carlo approximation to the oracle is used in the reported controlled experiment.

### S1.3 Predictive-construction implementation

The five primary predictive constructions are organized by their information source. TAIL and EVENT-EMP use event-level observations. GEV, KDE, and EMP use the 24 observed monthly maxima. This distinction is retained because comparisons across the two groups combine differences in information resolution and distributional construction, whereas TAIL versus EVENT-EMP and comparisons within GEV/KDE/EMP provide same-broad-information contrasts.

#### Event-level constructions

TAIL.

TAIL receives event-level maxima with month identifiers. Specification selection uses the first 19 months for candidate fitting and the final five months for internal predictive validation. Candidate thresholds are the empirical event-level quantiles

$$0.90, \quad 0.925, \quad 0.95, \quad 0.975.$$

104 For a candidate numerical threshold  $u$ , exceedances are strict: only observations  $Z > u$  enter the excess sample

105 
$$Y = Z - u.$$

106 The excess law is fitted by GPD maximum likelihood with location fixed at zero.

107 Monthly exceedance counts are formed over all fitting months, including months with zero exceedances. If  $\bar{N}$  and  $s_N^2$  denote  
108 the empirical monthly count mean and variance, the count law is Poisson when the variance-to-mean ratio is at most 1.2 or when  
109 the empirical mean is nonpositive. Otherwise the negative-binomial method-of-moments parameters are

110 
$$r = \frac{\bar{N}^2}{s_N^2 - \bar{N}}, \quad p = \frac{r}{r + \bar{N}}.$$

111 The corresponding probability-generating functions are

112 
$$P_N(z) = \exp\{\lambda(z - 1)\}$$

113 for the Poisson case and

114 
$$P_N(z) = \left[ \frac{p}{1 - (1 - p)z} \right]^r$$

115 for the negative-binomial case.

116 Candidate screening requires at least 100 strict exceedances. Shape stability is assessed from 50 nonparametric excess-sample  
117 bootstrap fits, with at least 40 successful fits required and the resulting 95% shape-interval width not exceeding 0.8. The success  
118 requirement is 80% of the configured replicate count, with at least 40 fits in the primary experiment. The probability-transformed  
119 fitted tail must satisfy a Kolmogorov–Smirnov  $p$ -value threshold of 0.01. For a negative-shape candidate, the natural finite  
120 endpoint

121 
$$e_u = u - \frac{\hat{\sigma}_u}{\hat{\xi}}$$

122 must satisfy

123 
$$e_u \geq \max_i X_i - 10^{-9},$$

124 where  $X_i$  are the fitting-period event maxima in kW. Candidate fits must have a finite validation score and satisfy

125 
$$\hat{\xi} < 1,$$

126 which ensures a finite first moment.

127 For each fitted candidate, we simulate  $m = 2,000$  monthly maxima  $Z_1, \dots, Z_m$  on the fitted candidate's natural support, without  
128 artificial upper clipping, and estimate their density with a Gaussian kernel:

129 
$$\hat{f}_m(y) = \frac{1}{mh} \sum_{j=1}^m \phi\left(\frac{y - Z_j}{h}\right), \quad h = s_Z m^{-1/5},$$

130 where  $\phi$  is the standard normal density and  $s_Z$  is the sample standard deviation with denominator  $m - 1$ . This is Scott's bandwidth  
131 as implemented by `scipy.stats.gaussian_kde`. For the five held-out monthly maxima  $Y_1, \dots, Y_5$ , the score is

132 
$$\ell_{\text{val}} = \frac{1}{5} \sum_{i=1}^5 \log \left\{ \max \left( \hat{f}_m(Y_i), \epsilon_{\text{tiny}} \right) \right\}, \quad \epsilon_{\text{tiny}} = 2.2250738585072014 \times 10^{-308}.$$

133 A numerically constant predictive sample has an undefined score and is excluded. If  $\mathcal{P}$  denotes the candidates that pass the  
134 screening and finite-mean conditions, the selection score is

135 
$$\ell_{\text{val}} - 0.10 \left| \hat{\xi} - \text{median}_{j \in \mathcal{P}}(\hat{\xi}_j) \right|.$$

136 An exact tie is resolved toward the lower threshold quantile.

137 The 19/5 split is used only for specification selection. After the numerical threshold is selected, that threshold is held fixed  
138 and the GPD parameters, exceedance-count law, and empirical subthreshold component are reconstructed using all 24 months.

If no admissible finite-mean primary candidate is available but an eligible threshold has at least 50 exceedances, the procedure uses an exponential-tail fallback with

$$\xi = 0$$

and scale equal to the corresponding mean excess. If no such threshold is available, the monthly-maxima empirical distribution is used. An invalid all-history GPD refit likewise invokes the prescribed fallback branch.

For a final primary TAIL fit, let  $F_0$  denote the empirical CDF of the subthreshold monthly-maximum component, let  $G_u$  denote the fitted GPD excess CDF, and let  $P_N$  denote the fitted exceedance-count probability-generating function. The monthly-maximum CDF is

$$\hat{F}_{\text{TAIL}}(x) = \begin{cases} F_0(x)P_N(0), & x < u, \\ P_N\{G_u(x-u)\}, & x \geq u. \end{cases}$$

The final mean is evaluated as

$$\mu_{\hat{F}_{\text{TAIL}}} = H_{\hat{F}_{\text{TAIL}}}(0),$$

the stop-loss function is obtained by deterministic integration of the fitted survival function, and continuous quantiles are obtained by deterministic inversion. Thus the simulation used for internal candidate validation is not reused as the final representation of the tariff-relevant functionals.

### EVENT-EMP.

EVENT-EMP uses the same event-level history as TAIL but imposes no threshold or parametric tail/count model. Let  $\hat{G}_E$  denote the pooled empirical CDF of all event-level maxima in the 24-month history, and let  $N_m$  denote the observed total event count in month  $m$ . The induced monthly-maximum CDF is

$$\hat{F}_{\text{EE}}(x) = \frac{1}{24} \sum_{m=1}^{24} \hat{G}_E(x)^{N_m}.$$

This is a finite discrete distribution. Its probability masses are the jumps of  $\hat{F}_{\text{EE}}$ ; its mean and stop-loss function are evaluated exactly by finite sums, and its quantiles use the discrete left inverse. There is no dedicated fallback branch.

TAIL and EVENT-EMP therefore use the same broad event-level information source but different distributional constructions. TAIL adds threshold selection, a parametric excess law and a fitted exceedance-count law, whereas EVENT-EMP combines a pooled empirical event distribution with the empirical distribution of observed monthly event counts.

## Monthly-maxima constructions

### GEV.

The controlled-experiment GEV construction uses the 24 observed monthly maxima. It is fitted using L-moments/probability-weighted moments rather than maximum likelihood. L-skewness is inverted deterministically for the EVT shape parameter within

$$[-0.90, 0.90],$$

and the location and scale parameters follow from the associated L-moment equations.

A successful raw GEV fit is conditioned on the nonnegative part of its model-implied natural support. If more than 1% of the raw fitted GEV probability lies below zero, or if the L-moment fit or resulting parameters are invalid, the procedure uses the empirical monthly-maxima distribution. Successful parametric fits satisfy the finite-mean requirement. When the fitted EVT shape is negative, its model-implied finite upper endpoint is retained; otherwise the upper support is unbounded.

The conditioned GEV mean and stop-loss function are evaluated deterministically over this support, using one-dimensional integration where required. Quantiles use the conditioned inverse distribution. This GEV construction is distinct from the K-GEV procedure used in the external manufacturing-load analysis.

177 *KDE.*

178 KDE is fitted to the 24 monthly maxima  $M_1, \dots, M_n$ , with  $n = 24$ , using bandwidth

$$179 \quad h = 1.06 s n^{-1/5},$$

180 where  $s$  is the sample standard deviation. The Gaussian mixture is conditioned on nonnegative demand. If

$$181 \quad \tilde{F}(x) = \frac{1}{n} \sum_{i=1}^n \Phi\left(\frac{x - M_i}{h}\right)$$

182 denotes the unconditioned Gaussian-mixture CDF and

$$183 \quad Z_+ = 1 - \tilde{F}(0),$$

184 then

$$185 \quad \hat{F}_{\text{KDE}}(x) = \begin{cases} 0, & x < 0, \\ \frac{\tilde{F}(x) - \tilde{F}(0)}{Z_+}, & x \geq 0. \end{cases}$$

186 For  $t \geq 0$ , its stop-loss function is

$$187 \quad H_{\hat{F}_{\text{KDE}}}(t) = \frac{1}{nZ_+} \sum_{i=1}^n \left[ (M_i - t) \Phi\left(\frac{M_i - t}{h}\right) + h \phi\left(\frac{t - M_i}{h}\right) \right],$$

188 where  $\Phi$  and  $\phi$  are the standard normal CDF and PDF. Setting  $t = 0$  gives the mean. Quantiles are obtained by deterministic  
189 inversion of the conditioned mixture CDF. If fewer than two observations are available or the observed variability is numerically  
190 degenerate, the implementation uses the empirical distribution.

191 *EMP.*

192 EMP assigns empirical probability to the 24 observed monthly maxima, with duplicate support values aggregated. For support  
193 points  $x_j$  with empirical probabilities  $p_j$ ,

$$194 \quad \mu_{\hat{F}_{\text{EMP}}} = \sum_j p_j x_j,$$

195 and

$$196 \quad H_{\hat{F}_{\text{EMP}}}(t) = \sum_j p_j (x_j - t)_+.$$

197 Quantiles use the empirical left inverse. No predictive-bootstrap layer is introduced into the final tariff-functional evaluation.

## 198 **S1.4 From predictive laws to tariff actions**

199 Every fitted predictive law is passed through the same tariff-action map. For the evaluated policies,  $\kappa\alpha > 1$ , and the interior  
200 contract probability is

$$201 \quad p^* = 1 - \frac{1}{\kappa\alpha}.$$

202 For a fitted law  $\hat{F}$ , define the left generalized quantile

$$203 \quad \hat{q}_p = \inf\{x : \hat{F}(x) \geq p\}.$$

204 The final contract level is

$$205 \quad \hat{D} = \text{clip}\left(\frac{\hat{q}_{p^*}}{\alpha}, \delta S, S\right).$$

206 Thus lower and upper boundary solutions are handled directly through the feasible interval. No predictive Monte Carlo candidate  
207 search or local order-statistic candidate set is used.

208 The fitted optimized tariff-mode values are

$$209 \quad V_{\hat{F}}(\text{cap}) = r_c S,$$

$$210 \quad V_{\hat{F}}(\text{act}) = r_d \mu_{\hat{F}},$$

212 and

$$213 \quad V_{\hat{F}}(\text{con}) = r_d \left\{ \hat{D} + \kappa H_{\hat{F}}(\alpha \hat{D}) \right\}.$$

214 The selected tariff mode minimizes these deterministic mode values. Exact mode-value ties are resolved by the fixed lexicographic  
215 order

$$216 \quad \text{actual demand} \prec \text{capacity} \prec \text{contract demand}.$$

217 For discrete predictive laws, the left-quantile convention fixes the generalized-quantile representative. For continuous laws,  
218 quantiles are obtained directly or by deterministic root inversion.

219 Although

$$220 \quad V_F(\text{cap}) = r_c S$$

221 does not itself contain a distributional functional, selection of the capacity mode depends on its comparison with the estimated  
222 actual-demand and contract-demand mode values. Under contract demand, the complete action additionally contains  $D$ ;  
223 consequently, tariff-mode correctness and full-action optimality are separate endpoints.

## 224 **S1.5 Primary comparisons and high- $p^*$ tariff stress**

225 The complete nine-cell primary summaries, mode-versus-within-contract regret decomposition, and core paired contrasts are  
226 reported in the main article. Supplementary Table S12 gives all ten region-specific paired comparisons for the final 10,000-history  
227 primary run. It is a complete reference table; the ten contrasts are not treated as one confirmatory family.

228 The high- $p^*$  stress analysis evaluates three target quantiles over ten utilization cells, using 10,000 histories and five predictive  
229 methods. Figure S1 and Table S1 report the history-level regret distributions, exact-zero-regret rates, and same-information  
230 paired comparisons. This stress design is separate from the nine-cell primary analysis.

## 231 **S1.6 Functional-error associations and certificate-conditioned loss**

232 Figure S2 summarizes the within-method association of regret with absolute mean, stop-loss, and  $q_{0.99}$  relative errors, and  
233 the concentration of regret among the largest-loss decisions. Table S7 reports the paired functional-correlation contrasts with  
234 simultaneous max- $|T|$  intervals. Table S2 reports within-contract loss among decisions stratified by certificate status. The  
235 certificate is a sufficient condition; observations outside the certified set are not classified as violations.

## 236 **S1.7 Numerical bounds and fitting-support sensitivity**

237 The theoretical sufficient conditions in the main article are summarized separately from their numerical implementation. The  
238 numerical audit uses upper constructions for the relevant distributional perturbations and a common numerical full-action solver  
239 allowance. For the primitive-action perturbation, the reported numerical upper quantity is

$$240 \quad \varepsilon_{\text{num}} = \max \left\{ r_d \left| \hat{\mu} - \mu_F \right|, r_d \kappa \Delta_{H, \text{upper}} \right\}.$$

241 It is a numerical upper bound on primitive-action perturbation, not the mode-value error  $e_V$  and not an exact analytic supremum.  
242 The feasible-threshold stop-loss quantity uses a direct 513-node estimate and a numerical upper allowance based on the  
243 monotone-CDF envelope and half-grid spacing; it is a numerical upper bound on the feasible stop-loss perturbation.

244 The primitive numerical sufficient-condition check is

$$245 \quad 2\varepsilon_{\text{num}} + \zeta_{\text{full, num}} < \gamma_F.$$

246 For each non-oracle competitor  $m$ , the direct contrast-Wasserstein check is

$$247 \quad K_{m,m_F^*} W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m},$$

248 where  $K_{\text{cap,act}} = r_d$ ,  $K_{\text{cap,con}} = r_d \kappa$ , and  $K_{\text{act,con}} = r_d \max\{1, |1 - \kappa|\}$ . The mode-wise pairwise Wasserstein comparator instead uses

$$249 \quad (L_m + L_{m_F^*}) W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m},$$

250 for every non-oracle competitor, and the uniform Wasserstein check is

$$251 \quad 2LW_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_F, \quad L = r_d \max\{1, \kappa\}.$$

252 The numerical Wasserstein upper quantity is based on CDF bounds; the point estimates reported in Table S6 use numerical  
 253 quadrature. The solver allowance and distributional numerical quantities are floating-point implementation bounds without  
 254 directed-rounding or interval-arithmetic guarantees. Table S14 reports the observed implementation-audit results: the primitive-  
 255 perturbation check was satisfied by 26,975/450,000 (5.994444%); the direct contrast-Wasserstein check by 57,501/450,000  
 256 (12.778000%); the mode-wise pairwise-Wasserstein check by 26,393/450,000 (5.865111%); and the uniform-Wasserstein check  
 257 by 5,329/450,000 (1.184222%). No observed mode-selection violations occurred among rows satisfying any of these four  
 258 numerical checks.

#### 259 *Fitting-support sensitivity.*

260 The primary controlled analysis evaluates each fitted law on the estimator-specific support described in Section S1.3. To assess  
 261 sensitivity to finite fitting bounds, a paired analysis uses the first 1,000 primary histories and the same nine tariff cells. Three  
 262 training-derived upper bounds are considered:

$$263 \quad B_{\text{tr}}^{(10)} = 10 \max_{1 \leq m \leq 24} M_m, \quad B_{\text{tr}}^{(20)} = 20 \max_{1 \leq m \leq 24} M_m, \quad B_{\text{tr}}^{(50)} = 50 \max_{1 \leq m \leq 24} M_m,$$

264 where  $M_1, \dots, M_{24}$  are the observed training monthly maxima. Because the data-generating distribution is known in the controlled  
 265 experiment, the sensitivity analysis also includes the oracle-referenced diagnostic bound

$$266 \quad B_{\text{or}} = 10q_{0.999}.$$

267 This diagnostic specification is used only in the sensitivity analysis. The same histories and tariff cells are retained under every  
 268 support rule. The support treatment affects the TAIL and GEV constructions; EVENT-EMP, KDE, and EMP are unchanged by  
 269 construction.

270 Panels (d–f) of Figure S3 summarize the paired support-policy comparisons relative to the primary specification. Panels (d) and  
 271 (e) show cellwise paired mean changes in regret for TAIL and GEV, respectively, with one paired Monte Carlo standard error, and  
 272 panel (f) shows the percentage of TAIL tariff-mode decisions that change. Table S11 reports, for both TAIL and GEV, the paired  
 273 mean regret change, its paired standard error, and the percentage of changed tariff-mode decisions in each tariff cell. Across the  
 274 displayed support rules and cells, the changed-mode percentage does not exceed 0.20% for TAIL or 0.40% for GEV. The largest  
 275 absolute cellwise paired mean-regret changes are 0.062 and 0.726 thousand CNY/month for TAIL and GEV, respectively.

## 276 **S1.8 Parameter recovery and threshold-fixed bootstrap interval coverage**

277 The parameter-recovery experiment is separate from the 10,000-history primary comparison. It uses 200 histories at each of 12,  
 278 24, and 60 months under a Poisson–GPD mechanism with generating threshold 80,000 kW,

$$279 \quad \xi = 0.15, \quad \sigma = 12,000 \text{ kW},$$

280 and a Poisson rate of 30 tail events per month.

281 The threshold continues to be selected by the fitted procedure rather than fixed at the generating threshold. Scale recovery is  
 282 therefore evaluated relative to the threshold-specific scale implied by GPD threshold stability.

283 The mean absolute errors of the fitted shape parameter are

284 0.0699, 0.0578, 0.0343

285 for 12, 24, and 60 months, respectively. Figure S4 reports the complete shape-error, relative-scale-error, and relative- $q_{0.99}$  error  
286 distributions.

287 For each history of length  $n$  months, we construct nominal 95% nonparametric percentile intervals for the GPD shape and  
288 scale. The selected threshold  $\hat{v}$  is held fixed. We draw 100 bootstrap samples with replacement from the excesses in the first  
289  $\lfloor 0.8n \rfloor$  months, preserving the excess count, and fit a GPD with location fixed at zero to each sample. The interval endpoints are  
290 the 2.5th and 97.5th percentiles of the valid parameter estimates; at least 75 finite fits with positive scale are required. Shape  
291 coverage is evaluated against  $\xi = 0.15$ , and scale coverage against  $\sigma(\hat{v}) = 12000 + 0.15(\hat{v} - 80000)$ . Table S10 reports coverage  
292 across 200 histories and 95% Clopper–Pearson intervals for its Monte Carlo uncertainty. Thresholds vary across histories and  
293 remain fixed within each bootstrap calculation.

## 294 S1.9 Physical clipping identity and sensitivity

295 Let latent event-level demand values be  $X_i$ , and let physical clipping operate at level  $c$ . Clipping commutes with maximization:

$$296 \min \left( \max_i X_i, c \right) = \max_i \min(X_i, c).$$

297 Thus clipping individual events before monthly maximization yields the same clipped monthly maximum as clipping the  
298 corresponding original monthly maximum.

299 Let

$$300 M_c = \min(M, c)$$

301 and let  $F_c$  denote its distribution. Pointwise,

$$302 M_c = M - (M - c)_+,$$

303 so

$$304 \mathbb{E}[M_c] = \mathbb{E}[M] - H_F(c).$$

305 For every  $t \leq c$ ,

$$306 (\min(M, c) - t)_+ = (M - t)_+ - (M - c)_+,$$

307 and therefore

$$308 H_{F_c}(t) = H_F(t) - H_F(c), \quad t \leq c.$$

309 For  $t \geq c$ ,

$$310 H_{F_c}(t) = 0.$$

311 The equality case  $t = c$  is included in the first identity.

312 The clipping experiment uses four physical regimes:

$$\begin{aligned} &\text{UNCAPPED : } c = 10q_{0.999}, \\ &\text{DISTANT : } c = \frac{q_{0.999}}{0.85}, \\ 313 &\text{Q999 : } c = q_{0.999}, \\ &\text{BINDING : } c = \frac{q_{0.999}}{1.25}. \end{aligned}$$

314 “UNCAPPED” is an experimental label for a distant finite physical clipping level. Its value  $c = 10q_{0.999}$  defines a clipping regime  
315 only in this robustness experiment. The clipping level  $c$  is applied to generated demand and is distinct from the fitted-law support  
316 treatment used in the primary controlled analysis.

317 The experiment uses 12- and 24-month histories,

$$318 u \in \{0.62, 0.63, 0.64\},$$

and 200 histories for each history-length–clipping–utilization setting. TAIL, KDE, EMP, and the robustness-only GATE selector are retained. In this experiment, normalized regret is one-month regret divided by one monthly capacity fee; it is distinct from the primary normalized regret  $R_F/V_F^*$ .

GATE selects TAIL when its fit status is `PASS`, the fraction of reconstructed maxima exceeding the support bound is at most 0.01, and two consecutive available threshold candidates pass screening with shape estimates of the same sign. Candidates are ordered by their threshold quantiles 0.90, 0.925, 0.95, 0.975; the qualifying pair need not include the threshold ultimately selected by TAIL. Each member must have at least 100 exceedances, a nominal 95% bootstrap shape-interval width at most 0.8, a uniform-transform KS  $p$ -value at least 0.01, and a compatible endpoint when its shape is negative, as defined in S1.3. GATE otherwise selects KDE. Shape-width screening uses 50 bootstrap resamples here and 500 in the physical-pipeline experiment in S1.10, with at least 80% valid refits. The support bound is the clipping level  $c$  here and 120 MW in S1.10. Two zero shape estimates count as having the same sign.

Figure S5 reports paired TAIL-minus-baseline ECDFs after averaging the three utilization cells within history. Negative values favour TAIL. ECDFs are used because the interquartile ranges of all 16 plotted contrast groups are zero and boxplots would obscure their nonzero tails.

### S1.10 Physical-pipeline stress at $\alpha = 1.20$

The physical-pipeline experiment evaluates high-amplitude, high-frequency, and clustered-event mechanisms at  $\alpha = 1.20$ . Each method, training length, and tariff cell is evaluated on 200 paired histories, for 72,800 decision records. Figure S6 and Table S3 report the distributional and cell-aggregated results.

### S1.11 Joint negative-binomial–Weibull challenge

The joint misspecification challenge combines a negative-binomial event count with Weibull exceedances. Figure S7 shows the cellwise distribution of regret across the actual, contract, and capacity oracle regions. Table S4 gives the complete cell-by-method results, and Table S5 gives the full regional paired contrasts. This experiment assesses a joint count and severity misspecification and is not used as a replacement for the physical-pipeline experiment above.

### S1.12 Crossed tariff geometry without an artificial upper fitting bound

The crossed-grid analysis evaluates five methods over 19 utilization values at  $\alpha \in \{1.05, 1.10, 1.15, 1.20\}$  using 1,000 shared histories, for 380,000 decisions. Figure S8 shows mean normalized full-action regret and oracle-mode accuracy. The curves connect discrete evaluated tariff cells for visual comparison; no interpolation or invariant within-region ranking is implied.

### S1.13 External manufacturing-load design and predictive constructions

The external analysis uses the public South Korean manufacturing-factory electricity dataset described by Lee, Baek and Kim<sup>1</sup> and its associated Figshare release<sup>2</sup>. The source contains one-minute electricity-demand records for 10 manufacturing facilities from March through September 2019. All 10 facilities are retained for descriptive upper-tail summaries, whereas eight factories satisfy the completeness requirements for formal rolling predictive evaluation.

The formal series is reconstructed using fixed, non-overlapping 15-minute averages. Timestamps must be valid, duplicate timestamps are resolved before reconstruction, and demand values must be numeric and nonnegative. A reconstructed 15-minute window is retained only when all 15 constituent one-minute observations are available; no imputation is applied to the primary tail series. A factory-month is eligible when at least 95% of its reconstructed windows are valid.

The primary natural-tail analysis removes documented demand-response intervals and the subsequent 60-minute rebound period. The all-observations sensitivity analysis retains these intervals while otherwise preserving the same rolling evaluation.

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March–June  $\rightarrow$  July,

March–July  $\rightarrow$  August,

March–August → September.

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### 392 *K-GTAIL.*

393 K-GTAIL is a runs-declustered peaks-over-threshold construction fitted to the normalized fixed-15 training series. Candidate  
394 threshold quantiles are

395  $\{0.90, 0.925, 0.95, 0.975\},$

396 and candidate run lengths are

397  $\{30, 60, 120\}$  minutes.

398 Exceedances are strict. Exceedances separated by no more than the candidate run length belong to the same run cluster, and each  
399 cluster is represented by its maximum excess. GPD excesses are fitted by maximum likelihood with location fixed at zero.

400 Every threshold–run candidate is evaluated through the leave-one-training-month-out procedure above. Within a fold, candidate  
401 predictive performance is measured on the held-out training-month maximum, and fold scores are averaged before specification  
402 selection. The selected threshold–run specification is then refitted using all training months.

403 The final external tail reconstruction uses a Poisson cluster-count law with rate estimated from the training clusters. A  
404 bootstrapped subthreshold monthly component is combined with simulated Poisson cluster counts and GPD excess severities to  
405 form predictive monthly maxima. The final external predictive representation contains 50,000 monthly-maximum draws.

406 K-GTAIL also retains prespecified fit screening and fallback branches as part of the complete finite-sample procedure. If  
407 an admissible gated tail specification is unavailable, or if the selected all-training fit fails the required diagnostic checks, the  
408 procedure can use an exponential-tail fallback with  $\xi = 0$  or an empirical monthly-maxima fallback. These fallback predictions  
409 remain in the formal score evaluation rather than being removed conditionally on fit status.

410 After leave-one-training-month-out selection, the all-training K-GTAIL refit applies the final finite-sample gates directly.  
411 It requires at least 50 run clusters, a 95% parametric-bootstrap interval for the GPD shape with width no greater than 0.8, a  
412 parametric-bootstrap Kolmogorov–Smirnov (p)-value of at least 0.05, and compatibility of any negative-shape endpoint with  
413 the observed final peaks and the normalized support bound 10. The shape and KS quantities use 500 parametric bootstrap  
414 replicates; at least 80% successful refits are required for these quantities to be considered valid. Candidate selection is by the  
415 mean leave-one-training-month-out validation log score, with ties broken by the lower threshold quantile and then the shorter  
416 run length. Each fold uses 2,000 predictive draws and the Gaussian-kernel log score defined in S1.3, evaluated at its held-  
417 out monthly maximum. Both the draws and the observation are divided by the mean monthly maximum of that fold’s fitting  
418 months. If the predictive draws have numerically zero standard deviation, the scoring density is Gaussian with mean  $\bar{Z}$  and  
419 standard deviation  $\max(10^{-6}|\bar{Z}|, 10^{-8})$ ; the same density floor  $\epsilon_{\text{tiny}}$  applies. For a negative-shape final fit, the endpoint gate is  
420  $\max_i X_i \leq u - \hat{\sigma}_u \hat{\xi} \leq 10$ , with  $X_i$  the normalized final cluster peaks.

421 If the final gate fails, the implementation compares exponential-tail and empirical fallbacks using leave-one-training-month-  
422 out scores. The exponential fallback is used only when its score is at least as good as the empirical fallback and the final fit has  
423 at least 25 clusters; otherwise the empirical monthly-maxima fallback is used. A final fit with fewer than three excesses also uses  
424 the empirical fallback.

### 425 *K-UGPD.*

426 K-UGPD is an auxiliary ungated POT/GPD construction using the same high-frequency training source and candidate thresh-  
427 old/run framework. It retains the tail construction without the complete K-GTAIL screening/fallback layer and is reported only  
428 as an external supplementary comparator.

### 429 *K-GEV.*

430 K-GEV is the external block-maxima parametric comparator. It fits a GEV distribution by maximum likelihood using SciPy’s  
431 generalized-extreme- value implementation on the normalized training monthly maxima. At least six nondegenerate training  
432 maxima are required; the fitted shape, location, and scale must be finite and the scale must be positive. The fit is represented by  
433 50,000 predictive draws. A draw outside the nonnegative interval  $([0, 10])$  is counted in the truncation fraction; if this fraction  
434 exceeds 1%, if a nonfinite draw occurs, or if fitting fails, the procedure falls back to the empirical distribution of the normalized  
435 training monthly maxima. Otherwise the draws are clipped to  $([0, 10])$ . This external construction is distinct from the controlled  
436 L-moment/PWM GEV implementation; no controlled-experiment estimation algorithm is transferred to K-GEV.

437 *Predictive scores.*

438 Whole-distribution predictive performance is evaluated by the continuous ranked probability score<sup>3</sup>,

$$439 \quad \text{CRPS}(F, y) = \mathbb{E}|X - y| - \frac{1}{2} \mathbb{E}|X - X'|,$$

440 where  $X$  and  $X'$  are independent under  $F$ .

441 For quantile level  $\tau$ , pinball loss is

$$442 \quad \rho_\tau(y - q_\tau) = (y - q_\tau) [\tau - \mathbf{1}\{y - q_\tau < 0\}].$$

443 The formal quantile-specific endpoints are the losses at

$$444 \quad \tau = 0.95 \quad \text{and} \quad \tau = 0.99.$$

445 The upper-tail score  $\text{qwCRPS}_{0.90}$  is the factor-two trapezoidal integration of pinball loss over the prespecified upper-tail  
446 quantile grid

$$447 \quad \mathcal{Q} = \{0.900, 0.905, \dots, 0.995\}.$$

448 Writing  $q_j = \hat{F}^{-1}(\tau_j)$  and  $\rho_j = \rho_{\tau_j}(y - q_j)$  for the predictive quantile and pinball loss at  $\tau_j \in \mathcal{Q}$ , the implemented discrete score is

$$449 \quad \text{qwCRPS}_{0.90} = 2 \sum_{j=1}^{|\mathcal{Q}|-1} \frac{\tau_{j+1} - \tau_j}{2} (\rho_j + \rho_{j+1}).$$

450 The score is evaluated on this fixed grid and is not reinterpreted as an interchangeable full-distribution CRPS.

451 For each method  $b$  and quantile level  $\tau$ , empirical exceedance frequency is summarized by

$$452 \quad \hat{e}_{b,\tau} = \frac{1}{32} \sum_{f=1}^8 \sum_{o=1}^4 \mathbf{1}\{y_{f,o} > q_{b,f,o,\tau}\}.$$

453 The nominal exceedance frequency is  $1 - \tau$ . The mean squared- exceedance statistic satisfies

$$454 \quad \bar{B}_{b,\tau} = (1 - \tau)^2 + (2\tau - 1)\hat{e}_{b,\tau}.$$

455 It is therefore an affine summary of empirical exceedance frequency at method-specific thresholds. Calibration is described  
456 relative to the nominal frequency, while comparative predictive loss is evaluated using CRPS, upper-quantile-weighted pinball  
457 loss, and the two individual pinball losses.

458 For aggregate method comparisons, the four origin-specific scores are first averaged within eligible factory, and differences  
459 are paired across the eight factories. Table S9 reports paired mean differences with pointwise 95% paired  $t$  intervals using seven  
460 degrees of freedom. The external analysis evaluates predictive distributions; no observed tariff action or realized electricity bill is  
461 available in this dataset.

## 462 **S1.14 External predictive scores and fitted-branch summaries**

463 The external analysis uses eight eligible factories and four rolling origins per factory. Table S8 reports formal fit-status counts  
464 across the 32 factory–origin fits for each method. Fallback fits remain part of the evaluated procedures. Table S9 reports paired  
465 predictive-score contrasts between K-GTAIL and the listed baselines. The score-specific intervals use eight factory-level paired  
466 differences; score contrasts are descriptive and do not imply a single winner across predictive targets. No real tariff bills are  
467 observed in this dataset.

## 468 S1.15 Numerical solver allowance

469 For a fitted distribution  $\hat{F}$ , the contract-mode objective is

$$470 \quad C_{\hat{F},\text{con}}(D) = r_d \{D + \kappa H_{\hat{F}}(\alpha D)\}, \quad D \in \mathcal{D}_S = [\delta S, S].$$

471 At a numerical contract level  $\tilde{D}$ , its one-sided subgradients are

$$472 \quad g_-(\tilde{D}) = r_d \left\{ 1 - \kappa \alpha \Pr_{\hat{F}}(M \geq \alpha \tilde{D}) \right\}, \quad g_+(\tilde{D}) = r_d \left\{ 1 - \kappa \alpha \Pr_{\hat{F}}(M > \alpha \tilde{D}) \right\}.$$

473 A valid subgradient  $s \in [g_-(\tilde{D}), g_+(\tilde{D})]$  is chosen nearest to zero. Convexity gives, for every feasible  $D$ ,

$$474 \quad C_{\hat{F},\text{con}}(D) \geq C_{\hat{F},\text{con}}(\tilde{D}) + s(D - \tilde{D}).$$

475 Therefore the fitted-objective gap is bounded by

$$476 \quad B_s(\tilde{D}) = \max_{D \in [\delta S, S]} s(\tilde{D} - D) = \begin{cases} s(\tilde{D} - \delta S), & s \geq 0, \\ -s(S - \tilde{D}), & s < 0. \end{cases}$$

477 The numerical contract allowance adds a scale-based floating-point term,

$$478 \quad \zeta_{\text{con,num}} = B_s(\tilde{D}) + 512 \epsilon_{\text{mach}} \max \left\{ 1, r_d(\tilde{D} + \kappa \alpha \tilde{D}) \right\}.$$

479 Actual-demand and capacity modes are singleton actions. The full-action numerical allowance therefore combines the contract  
480 gap, saved-versus- recomputed objective-evaluation allowances, and any saved candidate selection gap. Writing  $\eta_{\text{sel}}$  for the  
481 objective-evaluation allowance of the selected mode,  $\eta_{\text{max}} = \max_{j \in \mathcal{M}} \eta_j$  for the largest mode allowance, and  $\theta$  for the nonnegative  
482 saved candidate-selection gap, the common implementation allowance is

$$483 \quad \zeta_{\text{full,num}} = \zeta_{\text{con,num}} + \eta_{\text{sel}} + \eta_{\text{max}} + \theta.$$

484 For each mode  $j$ , the numerical objective-evaluation allowance is

$$485 \quad \eta_j = \left| \hat{V}_{j,\text{recomputed}} - \hat{V}_{j,\text{saved}} \right| + 512 \epsilon_{\text{mach}} \max \left\{ 1, \left| \hat{V}_{j,\text{recomputed}} \right|, \left| \hat{V}_{j,\text{saved}} \right| \right\}.$$

486 The saved candidate-selection gap is

$$487 \quad \theta = \max \left\{ 0, \hat{V}_{\hat{m},\text{saved}} - \min_{j \in \mathcal{M}} \hat{V}_{j,\text{saved}} \right\}.$$

488 These terms use saved-versus-recomputed objective differences with a scale-based round-off term; they are numerical allowances,  
489 not proven floating-point error bounds. A 307-case implementation cross-check compared contract-objective excess with the  
490 fitted quantile-reference objective; all 307 cases satisfied the subgradient numerical bound up to the stated  $10^{-3}$  CNY comparison  
491 tolerance.

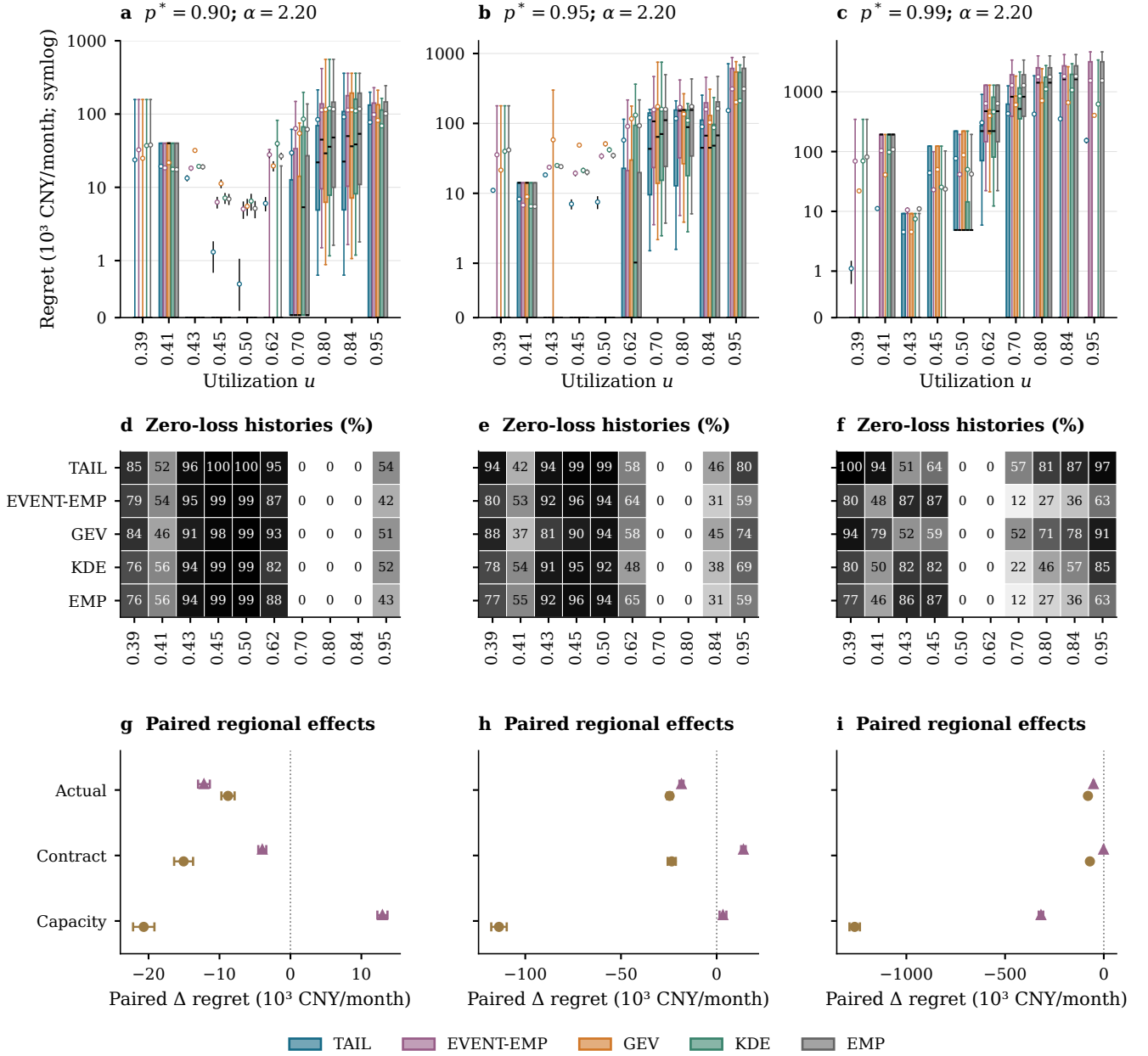
492 The mathematical subgradient inequality and the reduction from a valid contract objective-gap bound to a full-action allowance  
493 are exact. The floating-point evaluations used in the numerical implementation do not use directed rounding or interval arithmetic.  
494 Accordingly,  $\zeta_{\text{full,num}}$  is treated as a numerical solver allowance rather than an interval-certified error bound.

## 495 S1.16 Supplementary source and table index

496 The supplementary figure and table ownership is as follows.

- 497 • Figures S1–S8: high- $p^*$  tariff stress; functional-error and certificate diagnostics; numerical distribution-to-decision bounds
- 498 and fitting-support sensitivity; parameter recovery; physical clipping; physical-pipeline stress; joint NB–Weibull challenge;
- 499 and crossed tariff geometry, respectively.
- 500 • Table S1: high- $p^*$  paired comparisons;

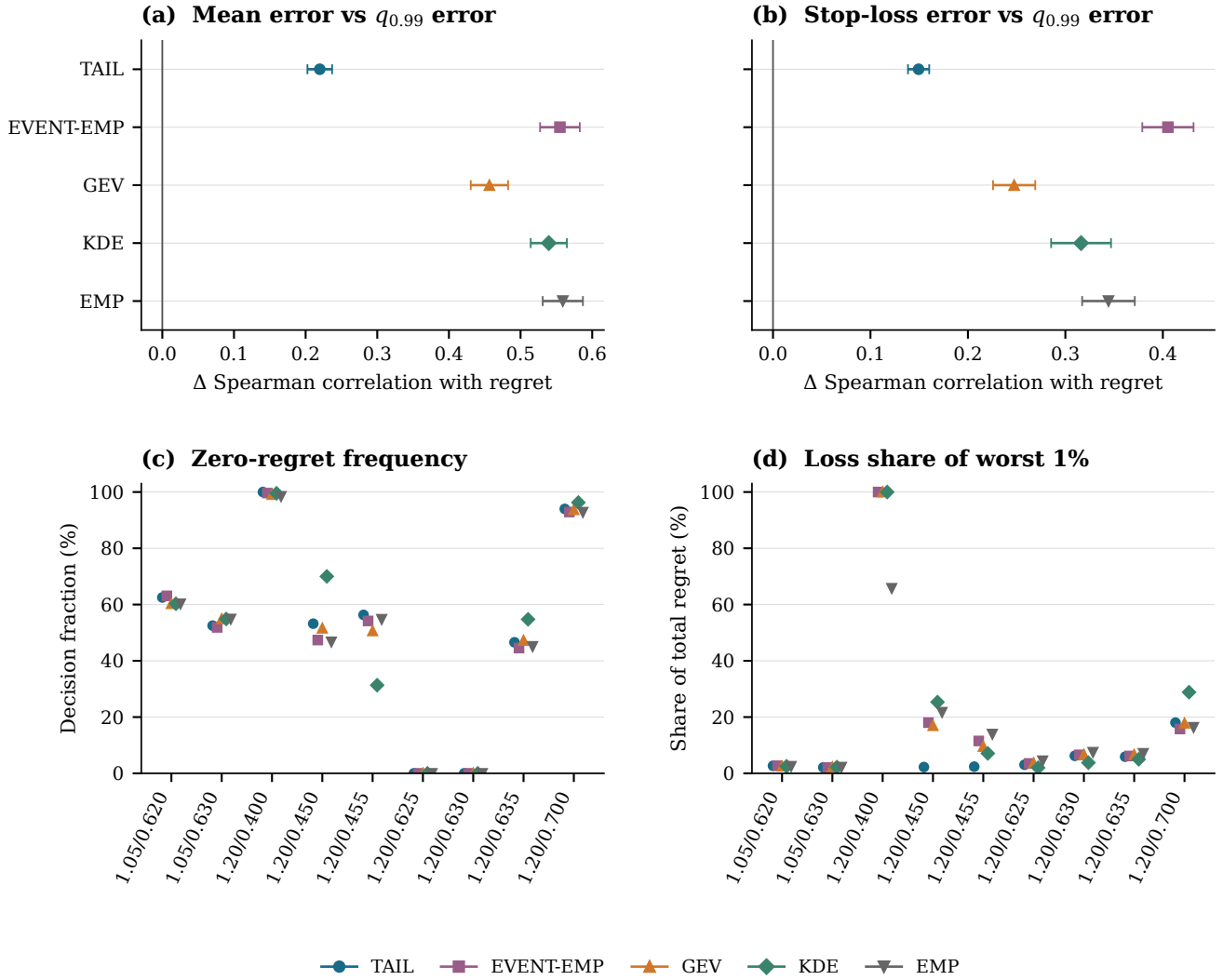
- 501 • Table S2: certificate-conditioned within-contract loss;
- 502 • Table S3: physical-pipeline stress;
- 503 • Tables S4–S5: joint NB–Weibull challenge;
- 504 • Table S6: numerical Wasserstein quantities;
- 505 • Table S7: functional-error contrasts;
- 506 • Table S8: external fit-status counts;
- 507 • Table S9: external predictive-score contrasts;
- 508 • Table S10: threshold-fixed bootstrap coverage;
- 509 • Table S11: fitting-support sensitivity;
- 510 • Table S12: primary regional paired contrasts;
- 511 • Table S13: primary tariff geometry and analytic demand-scale radii;
- 512 • Table S14: numerical sufficient-condition audit.



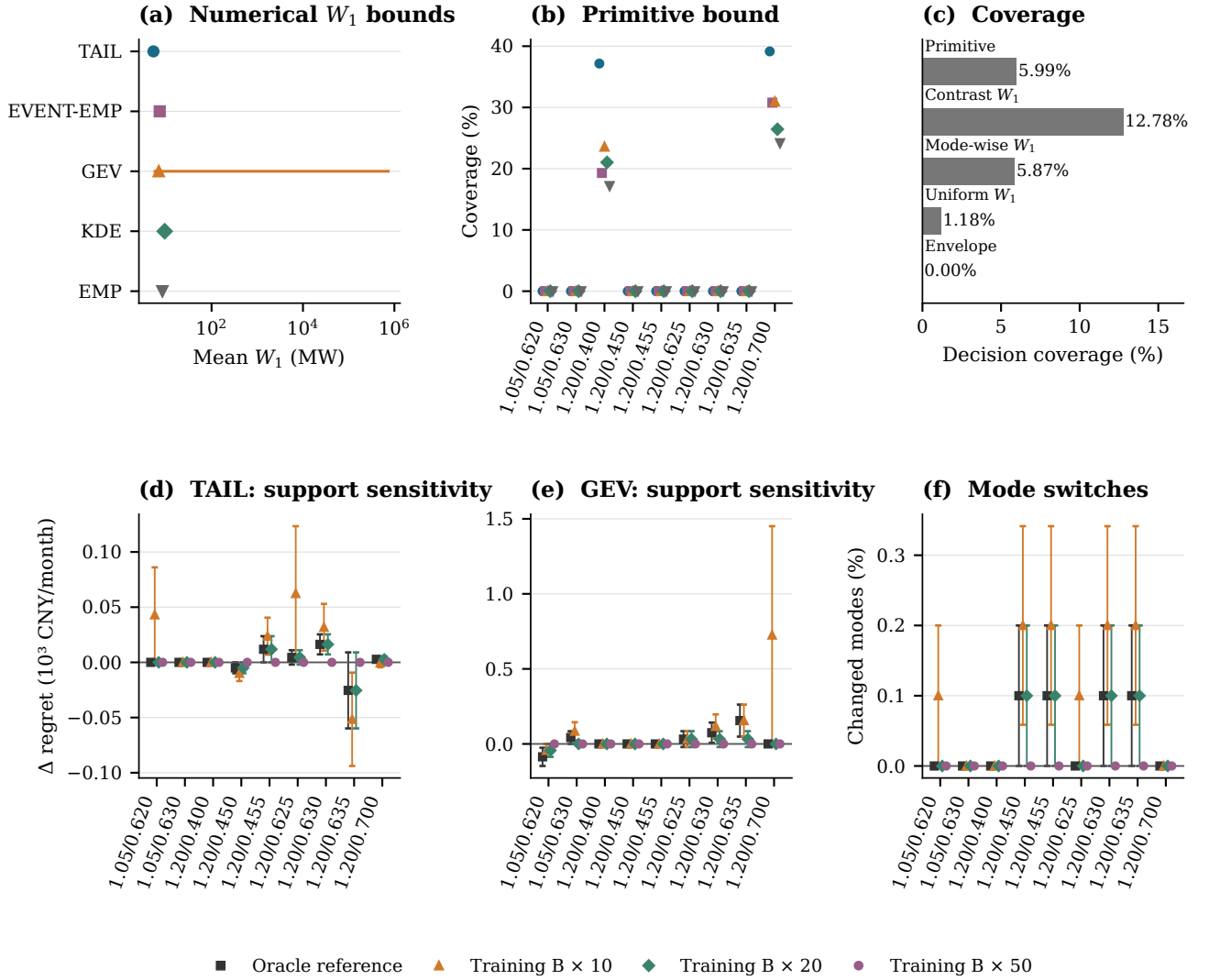
Boxes: P25–P75; whiskers: P10–P90; median tick; mean circle. 10,000 histories per method and cell.

Contrasts: brown circle TAIL – EVENT-EMP; purple triangle GEV – KDE. Mean and paired t95 CI.

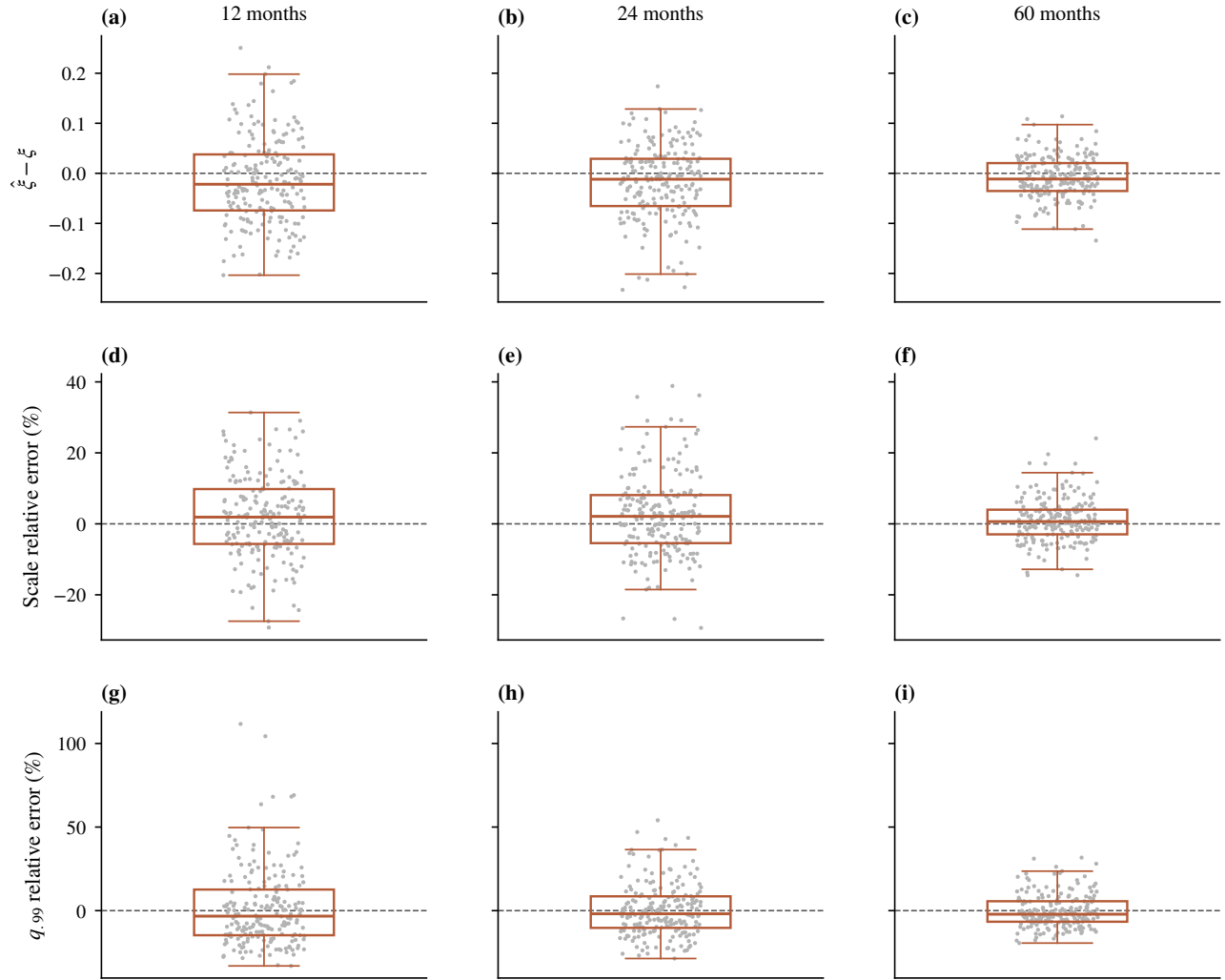
**FIGURE S1** High-contrast-quantile stress experiment at  $\alpha = 2.20$ , comprising 10,000 histories, five methods, three target quantiles, and ten utilization cells per target quantile (1,500,000 decisions). Columns correspond to target quantiles 0.90, 0.95, and 0.99. (a–c) Cellwise regret distributions: boxes span P25–P75, whiskers P10–P90, black ticks denote medians, and open circles denote means; black intervals at means are pointwise 95% Student t intervals (9,999 degrees of freedom). The symlog ordinate retains zero, is linear below one thousand CNY per month, and logarithmic above it; ranges vary by column. (d–f) Percentages of exactly zero regret, with the same grayscale mapping from 0 to 100 in all panels. Printed values are rounded to whole percentages; a printed 100 need not imply every history has zero regret. (g–i) Same-information paired regional mean differences: TAIL minus EVENT-EMP (circles) and GEV minus KDE (triangles). Cells are equally weighted within history and oracle region before differences are formed; intervals are paired t95 intervals. Negative values favour the first method. Data were reconstructed from saved fits without refitting or new simulation.



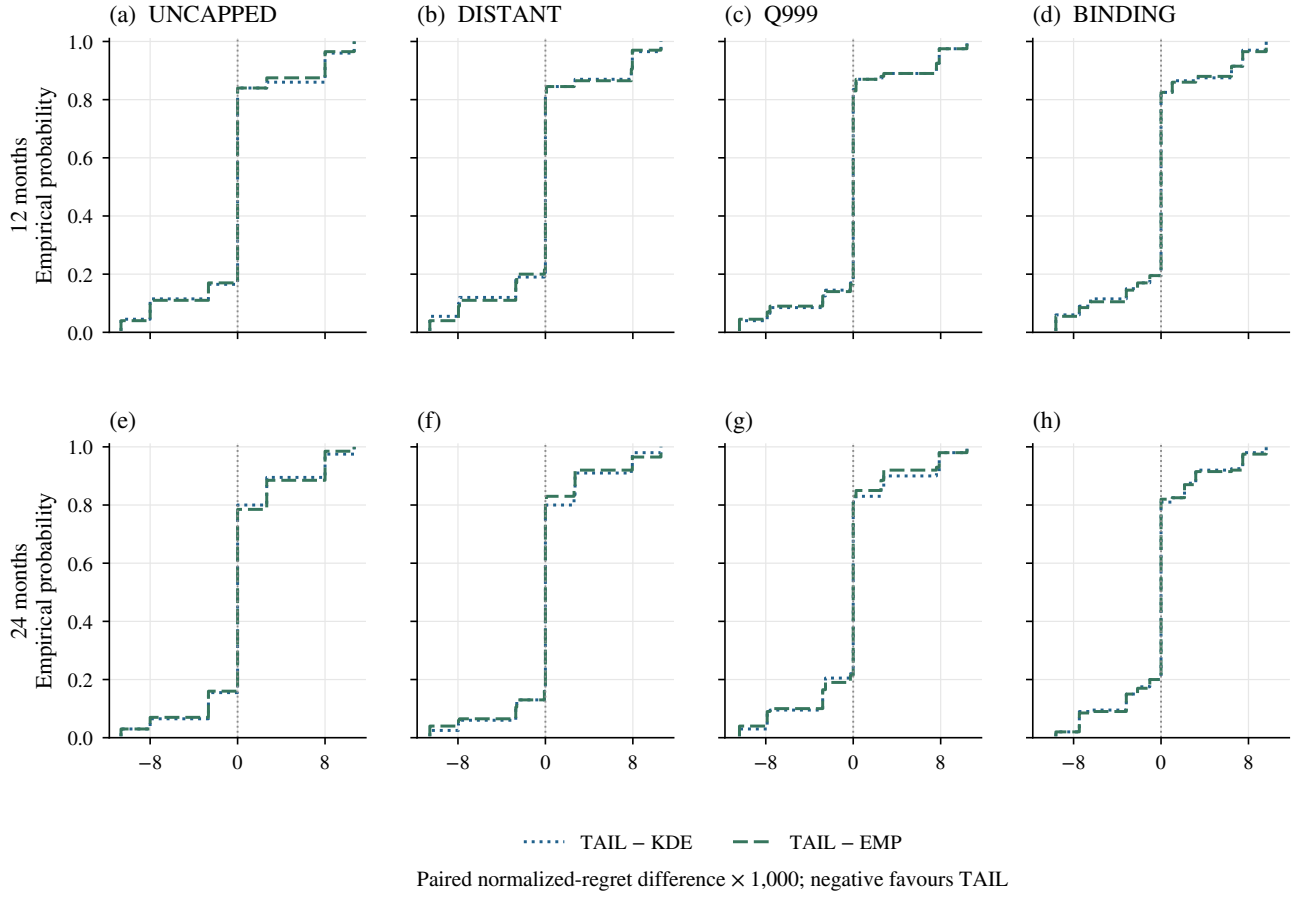
**FIGURE S2** Functional-error associations and loss concentration. (a,b) Within-method differences between the Spearman correlation of regret with absolute mean or stop-loss relative error and its correlation with absolute  $q_{0.99}$  relative error. Error and regret quantities are first averaged over nine cells within each of 10,000 histories. Intervals are the supplied simultaneous max- $|T|$  95% intervals for the family of ten correlation differences, based on 2,000 paired-history bootstrap resamples. They are not pointwise intervals. (c) Zero-regret frequency and (d) the share of aggregate regret accounted for by the worst 1% of decisions within each method–cell sample. The lower panels are descriptive summaries from 10,000 histories per method and cell. Offset, unconnected points distinguish methods at discrete tariff cells; the positions do not form a continuous tariff response.



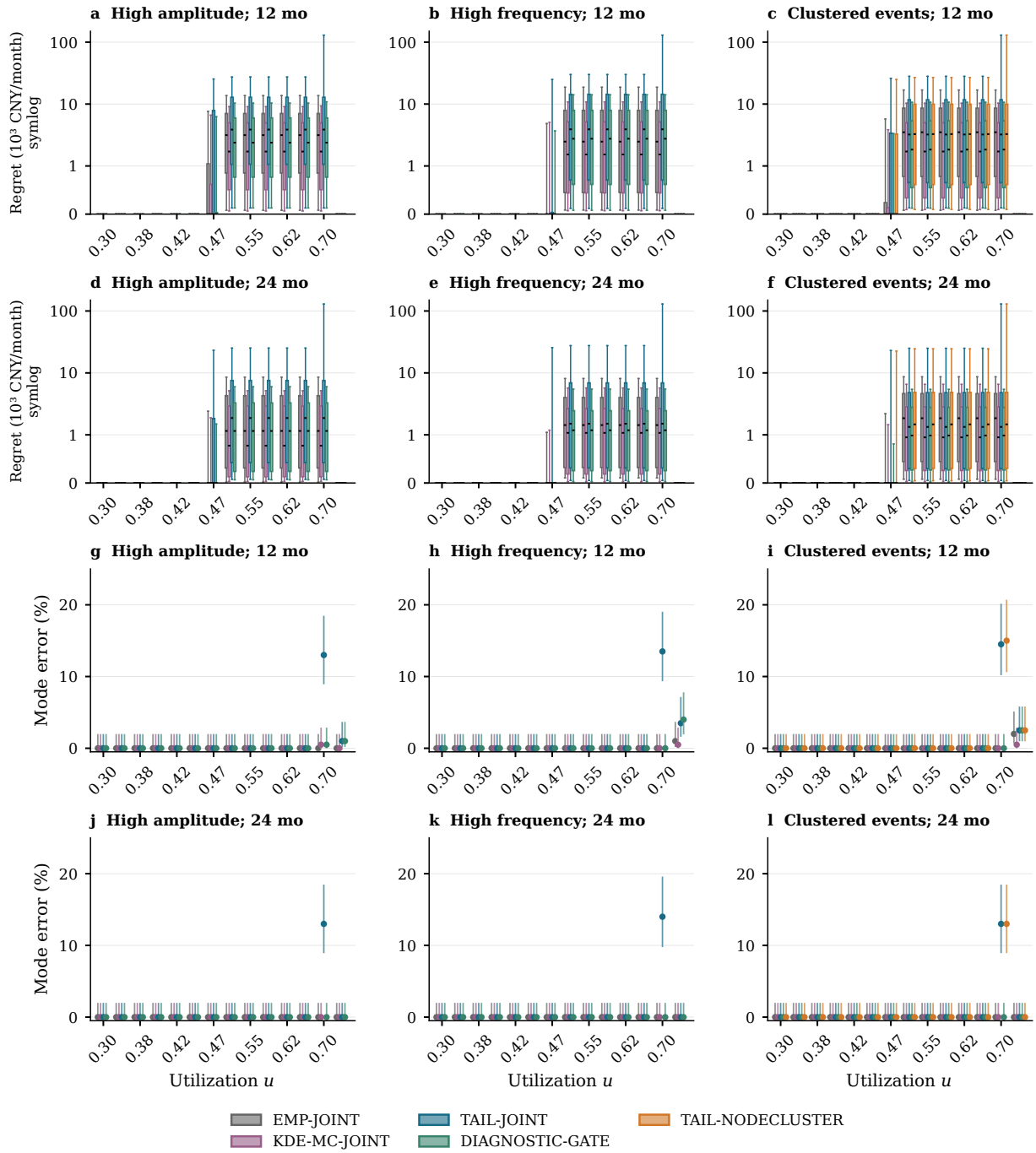
**FIGURE S3 Numerical distribution-to-decision bounds and fitting-support sensitivity.** (a) Mean  $W_1$  point estimates and CDF-based numerical lower and upper bounds over 10,000 fitted laws per method; the logarithmic axis retains the loose GEV upper bound. (b) Cellwise coverage of the primitive numerical sufficient-condition check by predictive method. (c) Overall coverage of the Primitive, contrast-Wasserstein, mode-wise pairwise-Wasserstein, uniform-Wasserstein, and coarse distributional-envelope checks over 450,000 primary decisions. The first four use the same numerical full-action solver allowance. The contrast-Wasserstein and mode-wise pairwise-Wasserstein checks differ in whether the common distributional perturbation is bounded directly for each tariff contrast or after separate mode-value bounds. (d,e) Paired mean regret changes for TAIL and GEV under alternative fitted-support rules relative to the primary specification. (f) Percentage of TAIL tariff-mode decisions that change under each support rule. Error bars in (d–f) are one paired Monte Carlo standard error over the first 1,000 primary histories. Training-derived support rules use multipliers 10, 20, and 50; Oracle reference denotes the diagnostic bound  $10q_{0.999}$ . Point estimates in (a), the CDF-based bounds, and the solver allowances are numerical rather than interval-certified quantities.



**FIGURE S4** Parameter-recovery experiment. Columns show training lengths of 12, 24, and 60 months; rows show GPD shape error, relative scale error, and relative  $q_{0.99}$  error. Each panel displays all 200 histories as grey points, with median, interquartile box, and 1.5-IQR whiskers. These are distribution summaries, not confidence intervals; the recovery experiment is separate from the primary analysis.



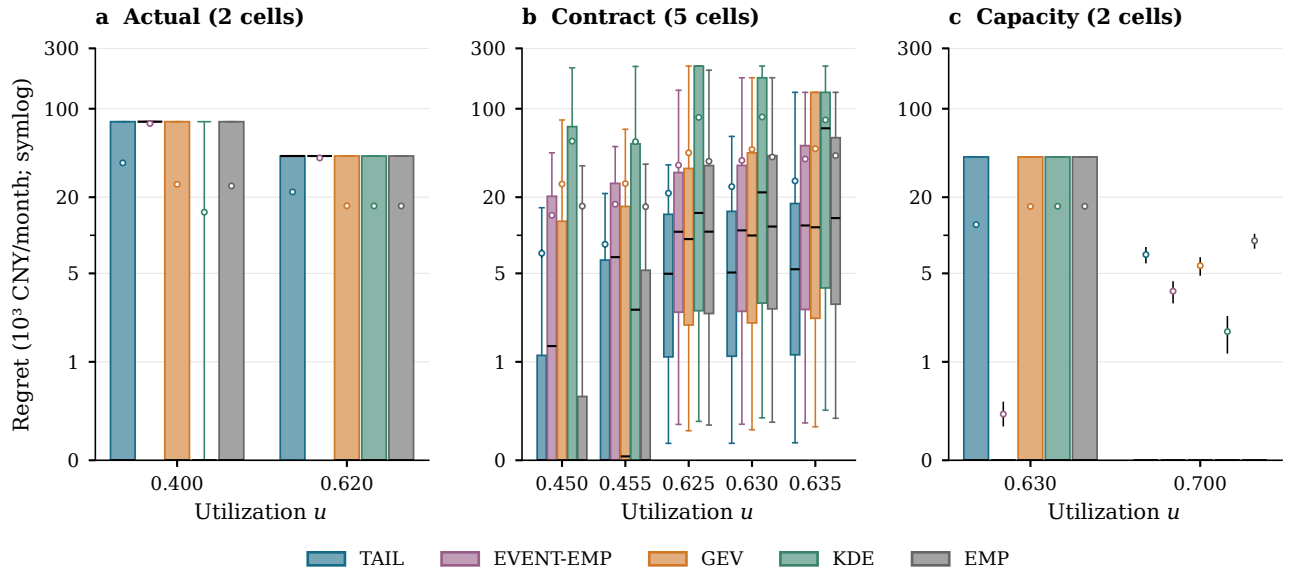
**FIGURE S5** Physical clipping sensitivity. Columns show the four frozen clipping regimes and rows show 12 or 24 training months. Each ECDF contains 200 paired history differences in normalized regret times 1,000; negative values favour TAIL. The full data range is shown, and the interquartile range is zero in all 16 groups. This descriptive robustness experiment is separate from the primary analysis.



200 histories per cell;  $\alpha = 1.20$ . Boxes: P25–P75; whiskers: P10–P90; median tick.

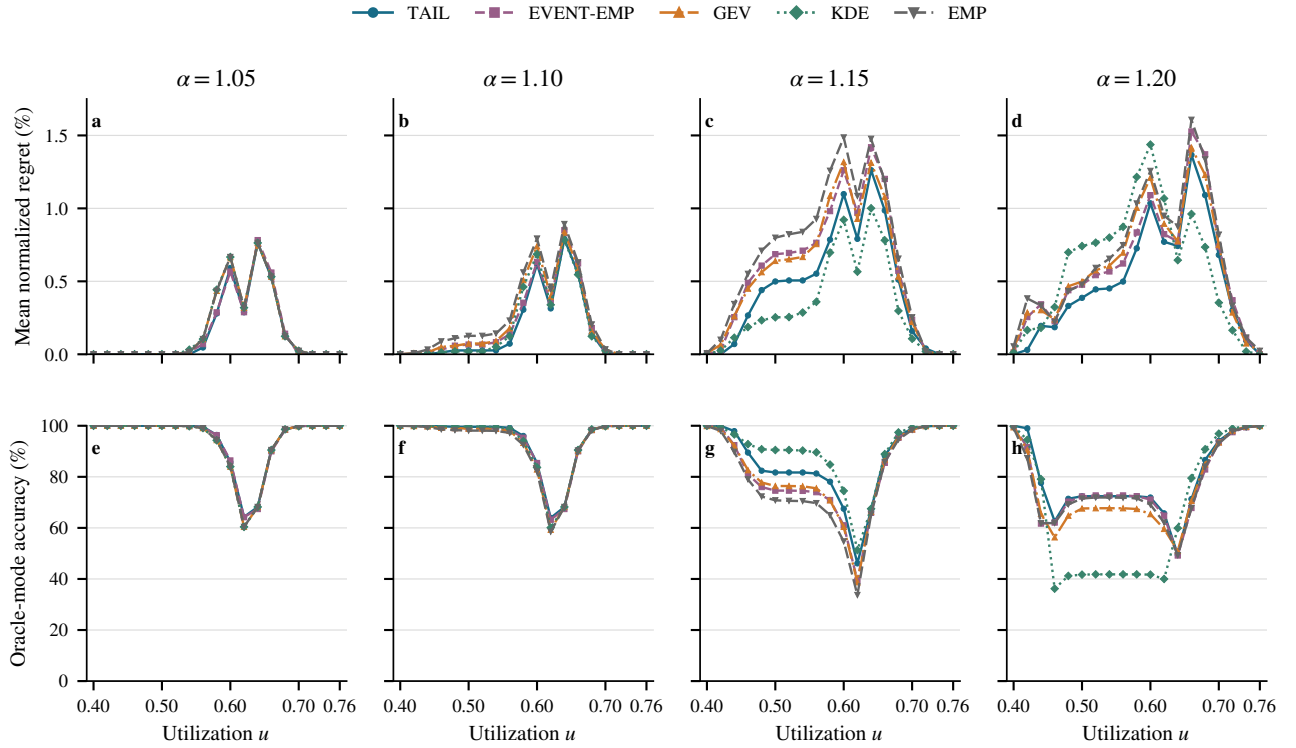
Lower panels: mode-error rates and Wilson 95% intervals; no interpolation between tariff cells.

**FIGURE S6** Physical-pipeline experiments at  $\alpha = 1.20$  using 200 paired histories per method, training length, and tariff cell (72,800 decisions). Columns represent high amplitude, high frequency, and clustered events. (a–f) Regret distributions at all 14 utilization cells for training lengths of 12 and 24 months: boxes span P25–P75, whiskers span P10–P90, and black ticks denote medians. The symlog ordinate retains zero and is linear below one thousand CNY per month. Outlying histories beyond the whiskers are not drawn individually; these are distribution summaries, not confidence intervals. (g–l) Mode-error proportions and pointwise Wilson 95% intervals at the same cells. Points are offset within each cell to separate methods and are not interpolated between cells. TAIL-NODECLUSTER applies only to the clustered-event scenario.



Boxes: P25–P75; black median; whiskers: P10–P90; open circle: mean. 10,000 histories per cell.

**FIGURE S7** Cellwise regret distributions under the negative-binomial–Weibull challenge, with 10,000 histories per method and cell. Panels group the two actual, five contract, and two capacity oracle cells; labels give utilization and cells are equally spaced. Filled boxes span P25–P75, black ticks denote medians, whiskers span P10–P90, and open circles denote means. Black intervals at means use the formal pointwise normal 95% approximation, mean plus or minus 1.96 standard errors. The symlog ordinate is linear below one thousand CNY per month and logarithmic above it. Histories outside the displayed quantile ranges are not drawn individually. Quantile ranges describe history variability and are distinct from uncertainty in the mean.



Each method-cell profile uses 1,000 shared histories; lines connect the 19 evaluated utilization cells.

**FIGURE S8** Complete five-method profiles over the final nonnegative, no-artificial-upper crossed tariff grid. Panels (a–d) show mean normalized full-action regret,  $R/V_F^*$ , and panels (e–h) show oracle-mode accuracy, for  $\alpha = 1.05, 1.10, 1.15$ , and  $1.20$ . Each panel covers the 19 evaluated utilization values; each method-cell estimate uses 1,000 shared histories, giving 380,000 decisions in total. Lines connect discrete evaluated cells for visual comparison and do not represent interpolation. The profiles are descriptive; no confidence intervals are shown and no invariant within-region ranking is asserted. TAIL and EVENT-EMP use event-level information; GEV, KDE, and EMP use monthly maxima.

**TABLE S1** Same-information comparisons under high contract quantiles.

| $p^*$ | First – second   | Oracle   | Cells | Difference | 95% CI               |
|-------|------------------|----------|-------|------------|----------------------|
| 0.90  | GEV – KDE        | Actual   | 1     | -12.19     | [-13.02, -11.36]     |
| 0.90  | GEV – KDE        | Capacity | 1     | +12.97     | [+12.26, +13.68]     |
| 0.90  | GEV – KDE        | Contract | 8     | -3.97      | [-4.53, -3.41]       |
| 0.90  | TAIL – EVENT-EMP | Actual   | 1     | -8.80      | [-9.73, -7.87]       |
| 0.90  | TAIL – EVENT-EMP | Capacity | 1     | -20.68     | [-22.18, -19.18]     |
| 0.90  | TAIL – EVENT-EMP | Contract | 8     | -15.06     | [-16.39, -13.72]     |
| 0.95  | GEV – KDE        | Actual   | 1     | -18.40     | [-19.46, -17.33]     |
| 0.95  | GEV – KDE        | Capacity | 2     | +3.28      | [+1.49, +5.07]       |
| 0.95  | GEV – KDE        | Contract | 7     | +13.97     | [+12.78, +15.15]     |
| 0.95  | TAIL – EVENT-EMP | Actual   | 1     | -24.59     | [-25.86, -23.31]     |
| 0.95  | TAIL – EVENT-EMP | Capacity | 2     | -113.74    | [-117.77, -109.70]   |
| 0.95  | TAIL – EVENT-EMP | Contract | 7     | -23.47     | [-25.59, -21.35]     |
| 0.99  | GEV – KDE        | Actual   | 2     | -52.28     | [-54.06, -50.49]     |
| 0.99  | GEV – KDE        | Capacity | 4     | -318.31    | [-329.72, -306.91]   |
| 0.99  | GEV – KDE        | Contract | 4     | -1.37      | [-3.10, +0.36]       |
| 0.99  | TAIL – EVENT-EMP | Actual   | 2     | -80.28     | [-82.37, -78.19]     |
| 0.99  | TAIL – EVENT-EMP | Capacity | 4     | -1261.71   | [-1288.14, -1235.28] |
| 0.99  | TAIL – EVENT-EMP | Contract | 4     | -70.45     | [-73.18, -67.71]     |

Loss differences are in  $10^3$  CNY/month. Each comparison uses the same 10,000 fitted primary histories and equal weights over the indicated region cells; intervals are pointwise paired  $t$  intervals. Alpha is 2.20. No multiple-testing p-values are used for interpretation.

## 513 Supplementary tables

**TABLE S2** Conditional within-contract regret by certificate status.

| Split    | Method    | Certified | Decisions | Histories | Mean  | 95% CI         |
|----------|-----------|-----------|-----------|-----------|-------|----------------|
| $d_{MV}$ | TAIL      | Yes       | 453       | 392       | 0.39  | [0.34, 0.45]   |
| $d_{MV}$ | TAIL      | No        | 17,142    | 6,812     | 7.81  | [7.48, 8.15]   |
| $d_{MV}$ | EVENT-EMP | Yes       | 284       | 254       | 7.14  | [5.86, 8.43]   |
| $d_{MV}$ | EVENT-EMP | No        | 17,581    | 6,948     | 11.73 | [11.26, 12.20] |
| $d_{MV}$ | GEV       | Yes       | 228       | 208       | 4.23  | [3.50, 4.95]   |
| $d_{MV}$ | GEV       | No        | 16,308    | 6,658     | 13.11 | [12.56, 13.65] |
| $d_{MV}$ | KDE       | Yes       | 144       | 131       | 12.19 | [10.42, 13.96] |
| $d_{MV}$ | KDE       | No        | 10,684    | 4,273     | 9.75  | [9.22, 10.28]  |
| $d_{MV}$ | EMP       | Yes       | 221       | 201       | 8.80  | [7.11, 10.48]  |
| $d_{MV}$ | EMP       | No        | 17,285    | 6,995     | 17.86 | [17.16, 18.56] |
| Pairwise | TAIL      | Yes       | 1,081     | 684       | 0.67  | [0.60, 0.73]   |
| Pairwise | TAIL      | No        | 16,514    | 6,601     | 8.02  | [7.68, 8.36]   |
| Pairwise | EVENT-EMP | Yes       | 714       | 465       | 7.61  | [6.64, 8.58]   |
| Pairwise | EVENT-EMP | No        | 17,151    | 6,813     | 11.73 | [11.25, 12.21] |
| Pairwise | GEV       | Yes       | 598       | 394       | 4.42  | [3.81, 5.02]   |
| Pairwise | GEV       | No        | 15,938    | 6,526     | 13.24 | [12.68, 13.79] |
| Pairwise | KDE       | Yes       | 408       | 262       | 11.93 | [10.71, 13.14] |
| Pairwise | KDE       | No        | 10,420    | 4,220     | 9.58  | [9.04, 10.11]  |
| Pairwise | EMP       | Yes       | 580       | 379       | 10.04 | [8.47, 11.61]  |
| Pairwise | EMP       | No        | 16,926    | 6,891     | 17.84 | [17.13, 18.55] |

The subset contains decisions for which both oracle and fitted-law selected modes are contract. The mean is the equally weighted mean of eligible history means, with a pointwise  $t$  interval using the displayed number of histories. Loss units are  $10^3$  CNY/month. Different subsets overlap across methods and certificate splits; their counts must not be added to obtain the number of independent histories.

**TABLE S3** Physical-mechanism stress tests at alpha 1.20.

| Scenario         | Months | Method           | Regret | MCSE  | Mode  | Within | Error (%) |
|------------------|--------|------------------|--------|-------|-------|--------|-----------|
| High amplitude   | 12     | GATE             | 2.114  | 0.236 | 0.110 | 2.005  | 0.107     |
| High amplitude   | 12     | EMP-JOINT        | 2.512  | 0.248 | 0.000 | 2.512  | 0.000     |
| High amplitude   | 12     | KDE-MC-JOINT     | 1.783  | 0.200 | 0.046 | 1.737  | 0.036     |
| High amplitude   | 12     | TAIL-JOINT       | 5.740  | 0.653 | 1.267 | 4.473  | 1.000     |
| High amplitude   | 24     | GATE             | 0.997  | 0.090 | 0.000 | 0.997  | 0.000     |
| High amplitude   | 24     | EMP-JOINT        | 1.283  | 0.117 | 0.000 | 1.283  | 0.000     |
| High amplitude   | 24     | KDE-MC-JOINT     | 0.842  | 0.079 | 0.000 | 0.842  | 0.000     |
| High amplitude   | 24     | TAIL-JOINT       | 4.451  | 0.534 | 1.204 | 3.247  | 0.929     |
| High frequency   | 12     | GATE             | 2.589  | 0.277 | 0.253 | 2.336  | 0.286     |
| High frequency   | 12     | EMP-JOINT        | 2.731  | 0.274 | 0.063 | 2.668  | 0.071     |
| High frequency   | 12     | KDE-MC-JOINT     | 1.784  | 0.185 | 0.032 | 1.752  | 0.036     |
| High frequency   | 12     | TAIL-JOINT       | 5.937  | 0.621 | 1.472 | 4.464  | 1.214     |
| High frequency   | 24     | GATE             | 0.947  | 0.090 | 0.000 | 0.947  | 0.000     |
| High frequency   | 24     | EMP-JOINT        | 1.286  | 0.116 | 0.000 | 1.286  | 0.000     |
| High frequency   | 24     | KDE-MC-JOINT     | 0.948  | 0.095 | 0.000 | 0.948  | 0.000     |
| High frequency   | 24     | TAIL-JOINT       | 4.311  | 0.544 | 1.297 | 3.014  | 1.000     |
| Clustered events | 12     | GATE             | 2.151  | 0.264 | 0.158 | 1.993  | 0.179     |
| Clustered events | 12     | EMP-JOINT        | 3.135  | 0.315 | 0.127 | 3.009  | 0.143     |
| Clustered events | 12     | KDE-MC-JOINT     | 1.855  | 0.217 | 0.032 | 1.823  | 0.036     |
| Clustered events | 12     | TAIL-JOINT       | 5.303  | 0.582 | 1.502 | 3.801  | 1.214     |
| Clustered events | 12     | TAIL-NODECLUSTER | 5.380  | 0.595 | 1.548 | 3.832  | 1.250     |
| Clustered events | 24     | GATE             | 0.936  | 0.093 | 0.000 | 0.936  | 0.000     |
| Clustered events | 24     | EMP-JOINT        | 1.572  | 0.145 | 0.000 | 1.572  | 0.000     |
| Clustered events | 24     | KDE-MC-JOINT     | 1.020  | 0.105 | 0.000 | 1.020  | 0.000     |
| Clustered events | 24     | TAIL-JOINT       | 3.782  | 0.513 | 1.205 | 2.577  | 0.929     |
| Clustered events | 24     | TAIL-NODECLUSTER | 3.768  | 0.510 | 1.205 | 2.563  | 0.929     |

Each scenario-by-training-length summary uses 200 independent histories, each evaluated at 14 equally weighted tariff cells (2,800 decision rows). GATE denotes DIAGNOSTIC-GATE. Losses are in  $10^3$  CNY/month. MCSE is calculated after averaging cells within history. The oracle is a fixed independent synthetic test pool.

**TABLE S4:** Cell-specific NB–Weibull challenge results.

| Region   | Alpha / u    | Method    | Mean regret | 95% CI         | Accuracy (%) |
|----------|--------------|-----------|-------------|----------------|--------------|
| actual   | 1.05 / 0.620 | EMP       | 17.05       | [16.64, 17.45] | 59.7         |
| capacity | 1.05 / 0.630 | EMP       | 16.96       | [16.56, 17.36] | 59.3         |
| actual   | 1.20 / 0.400 | EMP       | 24.55       | [23.83, 25.27] | 68.9         |
| contract | 1.20 / 0.450 | EMP       | 17.01       | [16.01, 18.01] | 93.9         |
| capacity | 1.20 / 0.700 | EMP       | 9.05        | [7.83, 10.27]  | 97.9         |
| contract | 1.20 / 0.635 | EMP       | 42.65       | [41.53, 43.77] | 81.3         |
| contract | 1.20 / 0.625 | EMP       | 38.49       | [37.23, 39.76] | 90.1         |
| contract | 1.20 / 0.630 | EMP       | 41.54       | [40.31, 42.77] | 86.2         |
| contract | 1.20 / 0.455 | EMP       | 16.82       | [15.86, 17.79] | 94.8         |
| actual   | 1.05 / 0.620 | EVENT-EMP | 40.83       | [40.64, 41.02] | 3.7          |
| capacity | 1.05 / 0.630 | EVENT-EMP | 0.47        | [0.34, 0.60]   | 99.0         |
| actual   | 1.20 / 0.400 | EVENT-EMP | 76.46       | [76.18, 76.73] | 3.2          |
| contract | 1.20 / 0.450 | EVENT-EMP | 14.39       | [13.91, 14.87] | 100.0        |
| capacity | 1.20 / 0.700 | EVENT-EMP | 3.62        | [2.90, 4.34]   | 99.0         |
| contract | 1.20 / 0.635 | EVENT-EMP | 40.05       | [39.01, 41.09] | 77.4         |
| contract | 1.20 / 0.625 | EVENT-EMP | 35.85       | [34.61, 37.08] | 90.0         |
| contract | 1.20 / 0.630 | EVENT-EMP | 39.05       | [37.86, 40.24] | 84.5         |
| contract | 1.20 / 0.455 | EVENT-EMP | 17.60       | [17.09, 18.10] | 100.0        |
| actual   | 1.05 / 0.620 | GEV       | 17.11       | [16.70, 17.52] | 59.6         |
| capacity | 1.05 / 0.630 | GEV       | 16.89       | [16.49, 17.29] | 59.4         |
| actual   | 1.20 / 0.400 | GEV       | 25.29       | [24.56, 26.01] | 68.0         |
| contract | 1.20 / 0.450 | GEV       | 25.41       | [24.24, 26.59] | 91.2         |
| capacity | 1.20 / 0.700 | GEV       | 5.75        | [4.79, 6.71]   | 98.6         |
| contract | 1.20 / 0.635 | GEV       | 48.33       | [47.07, 49.58] | 73.5         |
| contract | 1.20 / 0.625 | GEV       | 44.86       | [43.38, 46.34] | 84.6         |
| contract | 1.20 / 0.630 | GEV       | 47.69       | [46.29, 49.10] | 79.8         |
| contract | 1.20 / 0.455 | GEV       | 25.59       | [24.45, 26.74] | 92.2         |
| actual   | 1.05 / 0.620 | KDE       | 17.05       | [16.64, 17.45] | 59.7         |
| capacity | 1.05 / 0.630 | KDE       | 16.96       | [16.56, 17.36] | 59.3         |
| actual   | 1.20 / 0.400 | KDE       | 15.23       | [14.62, 15.84] | 80.7         |
| contract | 1.20 / 0.450 | KDE       | 55.44       | [53.71, 57.17] | 75.9         |
| capacity | 1.20 / 0.700 | KDE       | 1.73        | [1.16, 2.30]   | 99.6         |
| contract | 1.20 / 0.635 | KDE       | 81.62       | [80.10, 83.14] | 50.8         |
| contract | 1.20 / 0.625 | KDE       | 85.43       | [83.46, 87.40] | 63.7         |
| contract | 1.20 / 0.630 | KDE       | 86.07       | [84.29, 87.84] | 57.7         |
| contract | 1.20 / 0.455 | KDE       | 54.95       | [53.22, 56.67] | 77.5         |
| actual   | 1.05 / 0.620 | TAIL      | 22.02       | [21.61, 22.44] | 47.9         |
| capacity | 1.05 / 0.630 | TAIL      | 12.15       | [11.78, 12.52] | 70.8         |
| actual   | 1.20 / 0.400 | TAIL      | 37.34       | [36.56, 38.11] | 52.8         |
| contract | 1.20 / 0.450 | TAIL      | 7.21        | [6.66, 7.76]   | 98.3         |
| capacity | 1.20 / 0.700 | TAIL      | 7.04        | [5.99, 8.09]   | 98.3         |
| contract | 1.20 / 0.635 | TAIL      | 26.88       | [25.93, 27.83] | 85.8         |
| contract | 1.20 / 0.625 | TAIL      | 21.53       | [20.53, 22.53] | 94.1         |
| contract | 1.20 / 0.630 | TAIL      | 24.26       | [23.26, 25.26] | 90.9         |
| contract | 1.20 / 0.455 | TAIL      | 8.49        | [7.96, 9.03]   | 98.6         |

514 10,000 histories per method and cell. Losses are in  $10^3$  CNY/month; intervals use mean  $\pm 1.96$  SE. The oracle design has two actual, five contract and two capacity cells.

**TABLE S5:** Complete regional paired NB–Weibull contrasts.

| Region | First – second   | Mean difference | 95% CI           |
|--------|------------------|-----------------|------------------|
| actual | TAIL – EVENT-EMP | -28.97          | [-29.53, -28.40] |
| actual | TAIL – GEV       | +8.48           | [+7.98, +8.99]   |
| actual | TAIL – KDE       | +13.54          | [+13.04, +14.05] |
| actual | TAIL – EMP       | +8.88           | [+8.37, +9.39]   |
| actual | EVENT-EMP – GEV  | +37.45          | [+36.91, +37.99] |
| actual | EVENT-EMP – KDE  | +42.51          | [+42.03, +42.98] |
| actual | EVENT-EMP – EMP  | +37.85          | [+37.31, +38.38] |

| Region   | First – second   | Mean difference | 95% CI           |
|----------|------------------|-----------------|------------------|
| actual   | GEV – KDE        | +5.06           | [+4.80, +5.32]   |
| actual   | GEV – EMP        | +0.40           | [+0.25, +0.55]   |
| actual   | KDE – EMP        | -4.66           | [-4.91, -4.41]   |
| capacity | TAIL – EVENT-EMP | +7.55           | [+7.10, +8.00]   |
| capacity | TAIL – GEV       | -1.73           | [-2.19, -1.27]   |
| capacity | TAIL – KDE       | +0.25           | [-0.28, +0.78]   |
| capacity | TAIL – EMP       | -3.41           | [-3.91, -2.91]   |
| capacity | EVENT-EMP – GEV  | -9.28           | [-9.70, -8.86]   |
| capacity | EVENT-EMP – KDE  | -7.30           | [-7.74, -6.87]   |
| capacity | EVENT-EMP – EMP  | -10.96          | [-11.49, -10.44] |
| capacity | GEV – KDE        | +1.98           | [+1.58, +2.37]   |
| capacity | GEV – EMP        | -1.68           | [-2.04, -1.33]   |
| capacity | KDE – EMP        | -3.66           | [-4.20, -3.12]   |
| contract | TAIL – EVENT-EMP | -11.71          | [-12.37, -11.05] |
| contract | TAIL – GEV       | -20.70          | [-21.69, -19.71] |
| contract | TAIL – KDE       | -55.02          | [-56.50, -53.55] |
| contract | TAIL – EMP       | -13.63          | [-14.49, -12.76] |
| contract | EVENT-EMP – GEV  | -8.99           | [-9.99, -7.99]   |
| contract | EVENT-EMP – KDE  | -43.31          | [-44.77, -41.86] |
| contract | EVENT-EMP – EMP  | -1.91           | [-2.83, -1.00]   |
| contract | GEV – KDE        | -34.32          | [-35.51, -33.14] |
| contract | GEV – EMP        | +7.08           | [+6.46, +7.69]   |
| contract | KDE – EMP        | +41.40          | [+40.08, +42.71] |

515 Within each history, cells in the oracle region are equally weighted. There are 10,000 paired history differences per contrast. Units:  $10^3$  CNY/month; intervals: mean  $\pm 1.96$  paired SE.

516 Negative differences mean lower regret for the first method.

**TABLE S6:** Numerical Wasserstein brackets.

| Method    | Mean W1 (MW) | Mean lower (MW) | Mean upper (MW) |
|-----------|--------------|-----------------|-----------------|
| EMP       | 8.39         | 8.14            | 8.63            |
| EVENT-EMP | 7.34         | 7.12            | 7.55            |
| GEV       | 6.99         | 5.76            | 7.33e+05        |
| KDE       | 9.43         | 8.92            | 9.94            |
| TAIL      | 5.35         | 4.68            | 6.14            |

517 The  $W_1$  point estimates are 256-point Gauss–Legendre numerical quadrature estimates. The CDF-based lower and upper columns are numerical bounds, not sampling confidence

518 intervals or interval-certified enclosures. For GEV, the point estimate and CDF calculations use the fitted law conditioned on its nonnegative natural support.

**TABLE S7:** Functional-error correlation contrasts.

| Method    | Contrast        | Estimate | Simultaneous 95% CI |
|-----------|-----------------|----------|---------------------|
| TAIL      | Mean – q99      | +0.220   | [+0.203, +0.237]    |
| TAIL      | Stop-loss – q99 | +0.149   | [+0.139, +0.160]    |
| EVENT-EMP | Mean – q99      | +0.555   | [+0.527, +0.583]    |
| EVENT-EMP | Stop-loss – q99 | +0.405   | [+0.379, +0.432]    |
| GEV       | Mean – q99      | +0.457   | [+0.430, +0.483]    |
| GEV       | Stop-loss – q99 | +0.247   | [+0.226, +0.269]    |
| KDE       | Mean – q99      | +0.539   | [+0.514, +0.565]    |
| KDE       | Stop-loss – q99 | +0.316   | [+0.285, +0.347]    |
| EMP       | Mean – q99      | +0.559   | [+0.531, +0.587]    |
| EMP       | Stop-loss – q99 | +0.344   | [+0.317, +0.371]    |

519 The supplied max- $|T|$  intervals cover the family of ten within-method correlation differences. Resampling unit: history; 2,000 paired bootstrap resamples. Nine cells are averaged within

520 each of 10,000 histories.

TABLE S8: External fit-status counts, natural-tail variant.

| Method  | Formal fit status    | Factory-origin fits |
|---------|----------------------|---------------------|
| K-BLOCK | PASS                 | 32                  |
| K-EMP   | PASS                 | 32                  |
| K-GEV   | FALLBACK_EMPIRICAL   | 24                  |
| K-GEV   | PASS                 | 8                   |
| K-GTAIL | FALLBACK_EMPIRICAL   | 4                   |
| K-GTAIL | FALLBACK_EXPONENTIAL | 13                  |
| K-GTAIL | PASS                 | 15                  |
| K-KDE   | PASS                 | 32                  |
| K-POINT | PASS                 | 32                  |
| K-UGPD  | PASS                 | 32                  |

521 Eight factories and four rolling origins yield 32 fits per method. Labels are the formal stored status labels; fallback fits remain part of the evaluated pipeline.

TABLE S9: External predictive-score paired contrasts.

| Score       | Baseline | GTAIL – baseline | 95% CI             |
|-------------|----------|------------------|--------------------|
| crps        | K-BLOCK  | +5.627           | [-9.534, +20.789]  |
| crps        | K-EMP    | +3.005           | [-13.858, +19.869] |
| crps        | K-GEV    | +3.002           | [-14.318, +20.323] |
| crps        | K-KDE    | +4.654           | [-11.430, +20.737] |
| qwrps_090   | K-BLOCK  | -0.510           | [-1.895, +0.876]   |
| qwrps_090   | K-EMP    | -0.510           | [-1.895, +0.876]   |
| qwrps_090   | K-GEV    | -0.613           | [-2.177, +0.950]   |
| qwrps_090   | K-KDE    | -0.087           | [-0.938, +0.765]   |
| q95_pinball | K-BLOCK  | -2.594           | [-10.152, +4.964]  |
| q95_pinball | K-EMP    | -2.594           | [-10.152, +4.964]  |
| q95_pinball | K-GEV    | -3.268           | [-11.774, +5.239]  |
| q95_pinball | K-KDE    | -0.548           | [-5.256, +4.161]   |
| q99_pinball | K-BLOCK  | -4.071           | [-11.427, +3.285]  |
| q99_pinball | K-EMP    | -4.071           | [-11.427, +3.285]  |
| q99_pinball | K-GEV    | -4.481           | [-12.648, +3.687]  |
| q99_pinball | K-KDE    | -0.050           | [-2.356, +2.256]   |

522 Scores and intervals are multiplied by  $10^3$ . Each factory averages four origins; pointwise paired  $t$  intervals use eight factory differences and seven degrees of freedom. Score-specific  
523 comparisons are descriptive.

TABLE S10: Conditional parameter-interval coverage

| Months | Parameter | Covered / histories | Coverage (%) [95% MC CI] |
|--------|-----------|---------------------|--------------------------|
| 12     | xi        | 182 / 200           | 91.00 [86.15, 94.58]     |
| 12     | sigma     | 190 / 200           | 95.00 [91.00, 97.58]     |
| 24     | xi        | 185 / 200           | 92.50 [87.93, 95.74]     |
| 24     | sigma     | 182 / 200           | 91.00 [86.15, 94.58]     |
| 60     | xi        | 190 / 200           | 95.00 [91.00, 97.58]     |
| 60     | sigma     | 191 / 200           | 95.50 [91.63, 97.92]     |

524 Separate inherited recovery experiment, 200 histories per setting; not the primary experiment. Coverage conditions on the selected threshold. Clopper–Pearson intervals describe Monte  
525 Carlo uncertainty in coverage.

TABLE S11: Fitting-support policy sensitivity.

| Alpha / u  | Policy           | TAIL delta R | SE    | Mode (%) | GEV delta R | SE    | Mode (%) |
|------------|------------------|--------------|-------|----------|-------------|-------|----------|
| 1.05/0.620 | Oracle reference | +0.000       | 0.000 | 0.00     | -0.086      | 0.061 | 0.20     |
| 1.05/0.630 | Oracle reference | +0.000       | 0.000 | 0.00     | +0.042      | 0.042 | 0.10     |
| 1.20/0.400 | Oracle reference | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.450 | Oracle reference | -0.005       | 0.005 | 0.10     | -0.000      | 0.000 | 0.00     |
| 1.20/0.700 | Oracle reference | +0.003       | 0.003 | 0.00     | +0.000      | 0.000 | 0.00     |

| Alpha / u  | Policy           | TAIL delta R | SE    | Mode (%) | GEV delta R | SE    | Mode (%) |
|------------|------------------|--------------|-------|----------|-------------|-------|----------|
| 1.20/0.635 | Oracle reference | -0.025       | 0.034 | 0.10     | +0.155      | 0.107 | 0.40     |
| 1.20/0.625 | Oracle reference | +0.004       | 0.006 | 0.00     | +0.032      | 0.053 | 0.20     |
| 1.20/0.630 | Oracle reference | +0.016       | 0.009 | 0.10     | +0.074      | 0.068 | 0.30     |
| 1.20/0.455 | Oracle reference | +0.012       | 0.012 | 0.10     | -0.000      | 0.000 | 0.00     |
| 1.05/0.620 | Training x10     | +0.043       | 0.043 | 0.10     | -0.043      | 0.043 | 0.10     |
| 1.05/0.630 | Training x10     | +0.000       | 0.000 | 0.00     | +0.085      | 0.060 | 0.20     |
| 1.20/0.400 | Training x10     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.450 | Training x10     | -0.010       | 0.007 | 0.20     | -0.000      | 0.000 | 0.00     |
| 1.20/0.700 | Training x10     | -0.000       | 0.004 | 0.00     | +0.726      | 0.726 | 0.10     |
| 1.20/0.635 | Training x10     | -0.052       | 0.042 | 0.20     | +0.155      | 0.107 | 0.40     |
| 1.20/0.625 | Training x10     | +0.062       | 0.061 | 0.10     | +0.032      | 0.053 | 0.20     |
| 1.20/0.630 | Training x10     | +0.032       | 0.021 | 0.20     | +0.117      | 0.080 | 0.40     |
| 1.20/0.455 | Training x10     | +0.024       | 0.017 | 0.20     | -0.000      | 0.000 | 0.00     |
| 1.05/0.620 | Training x20     | +0.000       | 0.000 | 0.00     | -0.043      | 0.043 | 0.10     |
| 1.05/0.630 | Training x20     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.400 | Training x20     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.450 | Training x20     | -0.005       | 0.005 | 0.10     | -0.000      | 0.000 | 0.00     |
| 1.20/0.700 | Training x20     | +0.003       | 0.003 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.635 | Training x20     | -0.025       | 0.034 | 0.10     | +0.032      | 0.053 | 0.20     |
| 1.20/0.625 | Training x20     | +0.004       | 0.006 | 0.00     | +0.032      | 0.053 | 0.20     |
| 1.20/0.630 | Training x20     | +0.016       | 0.009 | 0.10     | +0.032      | 0.053 | 0.20     |
| 1.20/0.455 | Training x20     | +0.012       | 0.012 | 0.10     | -0.000      | 0.000 | 0.00     |
| 1.05/0.620 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.05/0.630 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.400 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.450 | Training x50     | +0.000       | 0.000 | 0.00     | -0.000      | 0.000 | 0.00     |
| 1.20/0.700 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.635 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.625 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.630 | Training x50     | +0.000       | 0.000 | 0.00     | +0.000      | 0.000 | 0.00     |
| 1.20/0.455 | Training x50     | +0.000       | 0.000 | 0.00     | -0.000      | 0.000 | 0.00     |

526 The TAIL and GEV column blocks each give paired mean change in regret, paired SE, and changed-mode percentage. All differences are relative to the primary support specification.

527 The first 1,000 paired histories form this separate support audit. Loss and SE units:  $10^3$  CNY/month. This subset does not replace the 10,000-history primary experiment.

**TABLE S12** Complete history-paired regret contrasts in the primary experiment.

| First – second   | Actual         | Contract       | Capacity       |
|------------------|----------------|----------------|----------------|
|                  | -0.61          | -1.68          | -2.67          |
| TAIL – EVENT-EMP | [-0.77, -0.44] | [-1.93, -1.43] | [-3.24, -2.10] |
|                  | +0.82          | -3.89          | +5.47          |
| GEV – KDE        | [+0.68, +0.97] | [-4.14, -3.63] | [+4.87, +6.07] |
|                  | -1.66          | +1.73          | -8.71          |
| KDE – EMP        | [-1.88, -1.43] | [+1.40, +2.06] | [-9.41, -8.00] |
|                  | -0.66          | -7.38          | +4.13          |
| TAIL – KDE       | [-0.85, -0.47] | [-7.69, -7.06] | [+3.44, +4.81] |
|                  | -1.48          | -3.49          | -1.34          |
| TAIL – GEV       | [-1.72, -1.24] | [-3.77, -3.22] | [-1.98, -0.70] |
|                  | -2.32          | -5.65          | -4.58          |
| TAIL – EMP       | [-2.62, -2.01] | [-5.99, -5.31] | [-5.26, -3.89] |
|                  | -0.84          | -2.16          | -3.23          |
| GEV – EMP        | [-1.01, -0.66] | [-2.37, -1.95] | [-3.62, -2.85] |
|                  | -0.88          | -1.81          | +1.33          |
| EVENT-EMP – GEV  | [-1.12, -0.63] | [-2.05, -1.57] | [+0.77, +1.89] |
|                  | -0.05          | -5.70          | +6.80          |
| EVENT-EMP – KDE  | [-0.26, +0.15] | [-5.99, -5.40] | [+6.10, +7.50] |
|                  | -1.71          | -3.97          | -1.90          |
| EVENT-EMP – EMP  | [-1.99, -1.43] | [-4.25, -3.68] | [-2.46, -1.35] |

First method minus second, in  $10^3$  CNY/month; negative values favour the first method. Brackets are pointwise paired  $t$  95% intervals from 10,000 equally weighted regional history means (9,999 degrees of freedom). The first two comparisons share event histories and monthly maxima, respectively; the remaining comparisons provide descriptive context. Intervals are not simultaneous and are not interpreted as family-wise evidence.

**TABLE S13** Primary tariff geometry and analytic demand-scale stability radii.

| $\alpha$ | Utilization | Oracle mode | $V_F^*$<br>( $10^3$ CNY/month) | $\gamma_F$<br>( $10^3$ CNY/month) | Relative gap<br>(%) | $\rho_F^{\text{contrast}}$<br>(kW) | $\rho_F^{\text{pair}}$<br>(kW) | $\rho_F$<br>(kW) |
|----------|-------------|-------------|--------------------------------|-----------------------------------|---------------------|------------------------------------|--------------------------------|------------------|
| 1.05     | 0.620       | Actual      | 5,336.473                      | 43.036                            | 0.806               | 1,195.45                           | 1,195.45                       | 298.86           |
| 1.05     | 0.630       | Capacity    | 5,294.120                      | 42.353                            | 0.800               | 1,176.47                           | 1,176.47                       | 294.12           |
| 1.20     | 0.400       | Actual      | 5,336.473                      | 318.649                           | 5.971               | 8,851.37                           | 2,950.46                       | 2,212.84         |
| 1.20     | 0.450       | Actual      | 5,336.473                      | 4.962                             | 0.093               | 137.84                             | 45.95                          | 34.46            |
| 1.20     | 0.455       | Contract    | 5,324.612                      | 11.861                            | 0.223               | 329.48                             | 109.83                         | 82.37            |
| 1.20     | 0.625       | Contract    | 5,276.065                      | 60.408                            | 1.145               | 839.00                             | 559.34                         | 419.50           |
| 1.20     | 0.630       | Contract    | 5,276.065                      | 18.055                            | 0.342               | 250.77                             | 250.77                         | 125.38           |
| 1.20     | 0.635       | Capacity    | 5,252.434                      | 23.631                            | 0.450               | 328.20                             | 328.20                         | 164.10           |
| 1.20     | 0.700       | Capacity    | 4,764.708                      | 511.357                           | 10.732              | 7,102.18                           | 7,102.18                       | 3,551.09         |

$V_F^*$  and  $\gamma_F$  are in thousand CNY/month; relative gap is  $100\gamma_F/V_F^*$ . The direct contrast radius is  $\rho_F^{\text{contrast}} = \min_{m \neq n} \gamma_{F,m}/K_{m,n}^*$ ; the mode-wise pairwise radius is  $\rho_F^{\text{pair}} = \min_{m \neq n} \gamma_{F,m}/(L_m + L_n^*)$ ; and the uniform radius is  $\rho_F = \gamma_F/(2L)$  with  $L = r_d \max\{1, \kappa\}$ . All are oracle-dependent analytic quantities in kW and contain no numerical-integration or solver allowance. For the primary coefficients,  $\kappa = 2$ , so  $K_{\text{act,con}} = 36$  CNY/kW compared with  $L_{\text{act}} + L_{\text{con}} = 108$  CNY/kW. Across these cells,  $\rho_F^{\text{contrast}} \geq \rho_F^{\text{pair}} \geq \rho_F$ .

**TABLE S14** Numerical sufficient-condition implementation audit.

| Condition                                      | Passing rows | Coverage (%) | Observed mode-selection violations |
|------------------------------------------------|--------------|--------------|------------------------------------|
| Primitive numerical check                      | 26,975       | 5.994444     | 0                                  |
| Contrast-Wasserstein numerical check           | 57,501       | 12.778000    | 0                                  |
| Mode-wise pairwise-Wasserstein numerical check | 26,393       | 5.865111     | 0                                  |
| Uniform-Wasserstein numerical check            | 5,329        | 1.184222     | 0                                  |

The primitive check is  $2\varepsilon_{\text{num}} + \zeta_{\text{full,num}} < \gamma_F$ . The contrast-Wasserstein check requires  $K_{m,m_F^*} W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m}$  for every competing mode, using the direct tariff-contrast constants defined in Section S1.7. The mode-wise pairwise check replaces  $K_{m,m_F^*}$  by  $L_m + L_{m_F^*}$ ; the uniform check is  $2LW_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_F$ , where  $L = r_d \max\{1, \kappa\}$ . All four use the same numerical full-action solver allowance. No observed mode-selection violations occurred among rows satisfying any of these checks. This is an implementation-audit result using floating-point numerical bounds, not an interval-certified numerical guarantee.

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## 528 AI-ASSISTED PREPARATION

529 OpenAI Codex was used to assist with code development and checking, analysis implementation, manuscript revision, language  
530 editing, and preparation of submission materials. The authors take responsibility for the content and interpretation of the work.

## 531 References

- 532 1. Lee E, Baek K, Kim J. Datasets on South Korean manufacturing factories' electricity consumption and demand response participation. *Scientific*  
533 *Data*. 2022;9:227. doi:10.1038/s41597-022-01357-8.
- 534 2. Lee E, Baek K, Kim J. Datasets on South Korean manufacturing factories' electricity consumption and demand response participation. Figshare.  
535 Version 9. 2022. doi:10.6084/m9.figshare.14822256.v9.
- 536 3. Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*. 2007;102:359–378.  
537 doi:10.1198/016214506000001437.